# Fundamentals of Goods Movement

*Course 1 — Introduction to Logistics and Goods Movement · Module 3 · Teaching content for the Theory Builder*

*Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.*

## Lesson 3.1 — The Journey of Goods from Supplier to Buyer

### Summary

This lesson explains the stages involved in moving goods from their point of origin to the final buyer or customer. Learners will explore how goods are prepared, collected, transported, handled, stored, and delivered, as well as the responsibilities of different logistics participants throughout the journey.

### Learning objectives

- Describe the typical stages involved in moving goods from a supplier to a buyer.
- Identify the main participants involved in the movement of goods.
- Explain how goods are prepared and collected for transportation.
- Describe the role of transportation, storage, and cargo handling during shipment movement.
- Explain the importance of shipment tracking and communication between logistics participants.
- Identify common factors that may affect the timely delivery of goods.
- Illustrate the movement of goods from origin to destination using a simple logistics scenario.

### Key terms

**Origin —** The location where a shipment begins its transportation journey.

**Destination —** The location where goods are intended to be delivered.

**Consignor —** The person or organisation that sends goods to another party.

**Consignee —** The person or organisation designated to receive goods.

**Shipment —** A quantity of goods transported together under a particular transport arrangement.

**Cargo Movement —** The physical transportation and handling of goods between locations.

**Collection —** The process of picking up goods from a supplier, warehouse, or designated location for transportation.

**Delivery —** The transfer of goods to the intended recipient or designated destination.

**Shipment Tracking —** The process of monitoring the location and progress of goods during transportation.

**Proof of Delivery (POD) —** A document or electronic record confirming that goods have been delivered or received.

## Teaching content

**KEY CONCEPT**

### The journey of goods as a chain of stages

When a buyer agrees to purchase goods from a supplier, the agreement is only the beginning. The sale is complete in practical terms only when the goods reach the buyer in the agreed quantity and condition. Everything that happens between the two premises is the journey of goods, the subject of this lesson.

Two terms locate the journey. The origin is the location where a shipment begins its transportation journey; the destination is the location where goods are intended to be delivered. A shipment is a quantity of goods transported together under a particular transport arrangement, whether a single pallet, a full truck, or a container holding thousands of cartons. The physical transportation and handling of goods between locations is called cargo movement.

An international shipment typically passes through nine stages.

1. Order and preparation. The buyer's order is confirmed, stock is verified, and the goods are selected, counted, and made ready.

2. Packing and stuffing. Goods are packed to protect them for the whole journey, then loaded into a container or vehicle.

3. Collection and inland movement to the export point. Goods are collected and moved by road or rail to a port, airport, or border crossing.

4. Export formalities. The exporting country's customs requirements are completed and the goods are cleared to leave.

5. The main international carriage. Goods travel between the two countries by sea, air, road, or rail.

6. Arrival and import formalities. Goods arrive in the importing country, where customs formalities are completed and release is granted.

7. Deconsolidation at the destination facility. Combined loads are separated so that each receiver's goods can be recovered.

8. Inland delivery. Goods move from the arrival point to the buyer's premises.

9. Receipt and verification. The buyer counts and inspects the goods and confirms delivery.

A domestic journey follows the same skeleton without the border stages.

One idea unites all nine stages: the handoff. At each stage boundary, responsibility for the goods passes from one party to another, from supplier to carrier, carrier to terminal, and finally carrier to buyer. Every handoff is a point where goods, information, and money must stay aligned, echoing the coordination of goods, information, and resources that this course identified as the heart of logistics. When a handoff fails, goods stop, records disagree, and payments are disputed. The rest of this lesson walks through the stages, and the handoffs between them.

**KEY CONCEPT**

## The participants who carry out the journey

No single organisation performs the whole journey. The previous module introduced the main participants in the logistics ecosystem; this lesson places them along the route. Two terms identify the sender and the receiver.

The consignor is the person or organisation that sends goods to another party. In a typical sale, the consignor is the seller or exporter: the farm cooperative dispatching pineapples, the factory dispatching cartons of tomato paste. The consignee is the person or organisation designated to receive the goods. In a typical sale, the consignee is the buyer or importer named in the transport documents. Usually the two names correspond to seller and buyer, but not always: a supplier may ship goods to a shop on the buyer's behalf, and the documents then name that receiving party as consignee. The names follow who sends and who is designated to receive, not who pays or who owns the vehicles.

Between them stand the supporting roles, each active at particular stages.

1. Carriers move the goods by sea, air, road, or rail.

2. Freight forwarders arrange and coordinate shipments on behalf of exporters and importers, booking carriage and organising handoffs.

3. Warehouse and terminal operators receive, store, consolidate, and hand over cargo at the facilities along the route.

4. Customs representatives prepare declarations and guide goods through the formalities of the exporting and importing countries.

5. Drivers, equipment operators, and terminal staff perform the physical cargo handling at each handoff.

Each participant was described in the previous module; what matters here is where each one acts. A single international journey may pass through a dozen organisations, and the consignor and consignee must be able to rely on all of them. That reliance holds only when every participant works from accurate, current information, which is why information flows matter in this lesson as much as the movement of the goods.

**KEY CONCEPT**

### Preparing and packing goods at the origin

The journey begins before anything moves. Stage one, order and preparation, starts when the buyer's order is received and confirmed. The supplier verifies availability in the ordered quantity, then warehouse staff pick, count, inspect, and stage the goods for dispatch. Errors caught here cost little; the same errors discovered at the destination cost money, time, and customer confidence. Preparation also includes checking that any certificates the goods require can be obtained before departure.

Packing decisions follow. Packaging serves two purposes across the entire journey: protection and handling. For protection, goods must survive stacking, mechanical handling, changes in humidity and temperature, and the vibration of long-distance transport, which is why packaging chosen for export journeys is usually sturdier than packaging for a short local delivery. For handling, the form of the packaging determines how goods can be lifted, stacked, and counted: cartons on pallets move by forklift in seconds, while loose cartons must be handled piece by piece.

Stage two is packing and stuffing. Stuffing is the industry term for loading goods into a container or vehicle. Goods are arranged and secured so that they cannot shift in transit, and a seal is applied to the container so that parties downstream can tell whether it was opened along the way. For perishable products, stuffing is where the cold chain begins: the unbroken path of temperature-controlled storage and transport, with continuous monitoring, that carries sensitive goods from origin to destination. Fresh produce is pre-cooled before it is loaded, and refrigerated equipment keeps it at the set temperature at every later stage. A consignment of fresh pineapples, for example, is packed into crates, stacked on pallets, and stuffed into a refrigerated container held at the required temperature; cartons of tomato paste are palletised and stuffed into a dry container. The packing choices made at the origin protect the goods through every handoff that follows.

**KEY CONCEPT**

### Collection and the inland movement to the export point

Stage three begins with collection. Collection is the process of picking up goods from a supplier, warehouse, or designated location for transportation. Collection is arranged in advance: the

transport operator, or the freight forwarder acting for the exporter, is given the collection address, the agreed time, and the details of the goods. The driver must arrive with instructions that match the cargo, because a mismatch between information and goods strands the cargo and disturbs every later connection.

From the supplier's premises, the goods move inland to the export point: the port, airport, or border crossing where the international carriage will begin. Two patterns are common. A full container load is often stuffed at the supplier's own premises: an empty container is delivered there, loaded and sealed, then trucked to the port. Smaller quantities are usually carried to a warehouse or container freight station, where the goods of several exporters are consolidated, meaning combined into one container or vehicle to share the cost of the journey. Consolidation saves money for shippers with small loads, but it adds an extra handling point and depends on accurate marking, because every consignment in the combined load must be identifiable so that it can be separated again at the destination.

The inland leg ends at the export point, and it ends against a clock. Vessels, aircraft, and scheduled road services depart at fixed times, and cargo must be handed over before the carrier's cut-off, the latest time at which cargo can be accepted for a particular departure. A shipment that misses its cut-off waits for the next departure, which may be days away. Careful exporters therefore plan the inland leg backwards from the cut-off, allowing time for the drive, the terminal queues, and the formalities before handover.

**KEY CONCEPT**

### Export formalities and the handover to the international carrier

Stage four takes place before the goods may leave the country. The customs formalities of the exporting country must be completed: an export declaration describing the goods, their value, and their destination is lodged with customs, and customs releases the goods for export when the declaration is in order. The exact requirements differ from country to country and from goods to goods. Some goods additionally require permits or certificates; plants and plant products, for instance, are commonly covered by a phytosanitary certificate issued by the exporting country's plant-protection authority. Responsibility for export formalities follows the trade terms of the sales contract, commonly expressed using the Incoterms 2020 rules; in most international sales the seller's side clears the goods for export.

When the formalities are complete, the goods are handed to the international carrier at the port, airport, or terminal. The carrier checks and accepts the cargo, and a transport document, for sea cargo a bill of lading, records what was received and on what terms it will be carried; the documents themselves belong to a later lesson. For containers, one safety requirement deserves mention here: before loading, the shipper must declare the container's verified gross mass, its actual total weight including the cargo, and a container whose weight has not been verified may not be loaded. Carriers also set a documentation cut-off alongside the cargo cut-off, so shipping information must be submitted on time as well as the physical boxes; a shipment missing either usually waits for the next vessel or flight.

The stage ends with loading and a significant handoff: the container is lifted aboard the vessel or the cargo onto the aircraft, and responsibility for the export leg has passed from the exporter's side to the carrier. A tracking event such as loaded or departed is recorded, opening the record of the main carriage that the consignor and consignee will follow from afar.

**KEY CONCEPT**

### The main international carriage

Stage five is the main carriage, sometimes called the line-haul: the long international leg between the export point and the arrival point in the destination country. The mode chosen, sea, air, road, or rail, depends on the nature of the goods, the distance, the urgency, and the cost; the next lesson, Modes of Transport: Sea, Air, Road, Rail, and Multimodal, compares the modes in detail, so they are only named here.

During the carriage, goods are in the continuous custody of the carrier, but they are not necessarily left untouched. Some journeys are direct, while others involve transhipment, meaning the transfer of the goods, or of the container holding them, from one vessel or aircraft to another at an intermediate hub. Every transfer is another cargo handling operation, another point at which delay or damage can occur, and another event in the tracking record. Goods may also be stored in transit, waiting for a connecting departure, space, or paperwork; storage during a journey is a normal part of many international movements.

The carrier's duties are to carry the goods with care, to follow the agreed route and schedule as far as conditions allow, and to keep the goods safe, with refrigerated equipment kept powered for sensitive cargo. The main carriage also concentrates the physical risks of the journey: weather, handling damage, and delay. Many shippers therefore arrange insurance cover for goods in transit; insurance is treated later in this course, so it is only noted here that the main carriage is the stage most often insured against.

Throughout the carriage, the participants report position and status events, departed, transhipped, expected arrival, arrived, which feed the tracking and communication practices described near the end of this lesson.

**KEY CONCEPT**

### Arrival, import formalities, and release

Stage six begins when the vessel berths, the aircraft lands, or the truck reaches the border crossing. For an import journey, arrival is not relief but the start of the most demanding formal stage, because the goods are now under the control of the importing country's customs authorities. Until the import formalities are completed and release is granted, they cannot lawfully be delivered to anyone.

The import formalities centre on a declaration. A declaration describing the goods, what they are, what they are worth, and where they come from, is lodged with the customs administration, usually by the importer or by a customs representative acting for the importer.

On the basis of the declaration, customs assesses the duties and taxes payable under the country's rules; rates and procedures differ from country to country and are a later subject of this course. Modern practice lodges much of this information before the goods arrive, known as pre-arrival declaration: when customs has the data in advance, it can analyse risk early and release goods much faster on arrival.

Customs does not inspect everything. Most administrations use risk-based selection: many, including Ghana's, sort declarations into channels, releasing some shipments on the documents alone, checking others' documents, and physically inspecting only shipments that present a risk. In Ghana, customs clearance is processed electronically through ICUMS, the national customs management system used at the country's ports of entry. When customs is satisfied, the declaration correct and any duties and taxes paid as applicable, the goods are released from customs control. Release is the gate between this stage and the rest of the journey: until it is granted, containers cannot leave the terminal and consignments cannot leave a customs facility. An error in the declaration discovered here leaves goods waiting exactly when they are closest to their destination, which is why accurate information at the arrival handoff matters as much as careful physical handling.

**KEY CONCEPT**

### Deconsolidation, destination storage, and collection

Stage seven is deconsolidation, the mirror image of the consolidation that may have happened at origin. Deconsolidation means separating a combined load into the individual shipments it contains, so that each consignee can receive its own goods. The work happens at a destination facility: a container freight station, a terminal, or a bonded warehouse. Where a container holds the goods of a single consignee, no separation is needed: the container is usually trucked intact to the consignee's premises and unpacked there. Where it holds the consolidated goods of several consignees, it is taken to a container freight station, unpacked, and its contents separated, checked against the documents, and held ready for each consignee. Each consignee's goods must be identified and recorded correctly, because a consignment placed in the wrong pile is one of the most confusing errors to unwind later.

Goods waiting at the destination, for release, for collection, or for onward transport, are stored at terminals, freight stations, or warehouses. Destination storage is a service with a clock attached. Terminals and carriers normally allow a limited period of free time after arrival or discharge, and once that period ends, storage and related charges begin to accumulate day by day; the lengths of these periods and the amounts charged are set by each carrier and terminal, not by any general rule. That clock is one of the practical reasons importers complete their formalities promptly and arrange collection early.

Stage seven closes with collection: once the goods are released, they are collected by road for the final inland delivery, usually by a truck arranged by the consignee or the consignee's representative. Importers who use tracking information to book the truck against the expected

release date save days at this handoff; those who wait until release is granted begin the search for transport while the free-time clock is already running.

KEY CONCEPT

### Final delivery, verification, and proof of delivery

Stage eight is the inland delivery, the movement of goods from the arrival point to the buyer's premises. This is the stage most people picture when they think of logistics, yet it is only one of nine, and by the time it happens, most of the journey's decisions have been made.

Stage nine closes the journey with delivery and verification. Delivery is the transfer of goods to the intended recipient or designated destination. Receiving goods well is a discipline of its own: the receiving staff count the packages against the accompanying documents, inspect their condition, and compare the goods with the order itself. Any discrepancy, a shortage, a damaged carton, a wrong item, is recorded at the moment of receipt, while the carrier is still present and responsibility can still be traced to the correct party. A discrepancy recorded a day later is far harder to investigate, because nobody can say when the loss or damage occurred.

The delivery is confirmed by proof of delivery, usually abbreviated to POD. A proof of delivery is a document or electronic record confirming that goods have been delivered or received. It may take the form of a signed delivery note, a photograph, or an electronic confirmation carrying the time, place, and receiver's identity. The POD performs three jobs in one: it closes the final handoff, because responsibility for the goods passes to the buyer; it anchors the records, because the buyer's stock records increase and the supplier's records show the order fulfilled; and it supports what follows commercially, since invoicing and payment under the agreed terms generally depend on evidence of what was actually delivered.

The journey, then, is complete not when the truck arrives, but when the goods are verified, the proof of delivery is issued, and the records on both sides agree. The flow of money catches up with the flow of goods according to the payment terms of the contract, ending with goods, information, and money reconciled.

KEY CONCEPT

### Shipment tracking, communication, and the causes of delay

Shipment tracking is the process of monitoring the location and progress of goods during transportation. Tracking begins at booking and ends at proof of delivery: an event is recorded at every meaningful point of the journey, collected, export cleared, loaded, departed, transhipped, arrived, released, delivered. The events come from the systems of carriers and terminals, from drivers' updates, and from forwarders' messages, letting every participant see where the shipment stands without seeing the goods.

Tracking matters for four reasons.

1. Planning. Parties prepare each stage before it arrives: trucks are booked, warehouse labour is scheduled, and cold-store space is reserved against the expected arrival, not after it.

2. Early detection. A problem seen early is a problem with options: documents can be corrected before arrival, an alternative connection can be sought, and customers can be warned while adjustment is still possible.

3. Coordination. Each organisation in the chain plans on facts instead of guesses, which reduces the small mismatches that waste time at every handoff.

4. Customer service. Businesses that can answer credibly where an order is and when it will arrive keep the confidence of their own customers even when the journey is imperfect.

Tracking is only half of the practice; communication is the other half. A tracking event creates value only when it reaches the parties who must act, in time to act. Communication between logistics participants is therefore deliberate: the parties agree who is told what, when, and through which channel, from the collection notice at origin to the proof of delivery at destination.

Even with good tracking and communication, delays occur. The common factors affecting timely delivery include the following.

1. Incomplete or incorrect documents, which hold goods at export or import formalities.

2. Congestion at ports, airports, borders, or on inland roads.

3. Missed cut-offs and missed connections, which push a shipment to the next departure.

4. Weather and seasonal conditions on sea, in the air, and on land.

5. Shortages of equipment or capacity, when no container, truck, or vessel space is available.

6. Inspections and holds by authorities during formalities.

7. Poor communication itself, when information arrives late, incomplete, or not at all.

Most of these causes cannot be eliminated by any single party, but all of them can be anticipated, monitored, and communicated, which is why tracking and communication are not extras added to a journey but core operations of it.

**WORKED EXAMPLE**

### An export journey: fresh pineapples from Ghana to Europe

Consider a farm cooperative in Ghana's Eastern Region that grows fresh pineapple for export. A supermarket distributor in Rotterdam orders a consignment for a fixed sailing date. The cooperative books the shipment through a freight forwarder; all quantities and times are illustrative.

Step 1: Order and preparation. The distributor's order specifies the variety, the quantity, 18 pallets, the quality standard, and the sailing date. The cooperative confirms availability and plans the harvest to match, so fruit is cut at the right maturity and moves to packing within hours.

Step 2: Packing. The fruit is graded, packed into crates, and the crates are stacked on pallets. Because pineapple is a plant product, the cooperative obtains a phytosanitary certificate from the national plant-protection authority in good time.

Step 3: Stuffing. A refrigerated container, a reefer, is delivered to the packing station and pre-cooled. The 18 pallets are stuffed, secured against movement, and the doors are sealed; temperature recording begins. The cold chain must now hold until Rotterdam.

Step 4: Inland movement. The reefer is trucked to Tema Port, timed to arrive before the shipping line's cargo cut-off for the booked vessel.

Step 5: Export formalities. The cooperative's agent lodges the export declaration, and the exporting country's customs completes the formalities, checking the declaration and accompanying certificates.

Step 6: Handover and loading. The terminal receives the container before the cut-off, the verified gross mass having been declared as required before loading, and the vessel loads the box, plugging it into its power supply.

Step 7: Main carriage. For this example, assume a sea voyage of about twelve days to Europe. The container is kept at temperature and powered throughout, and the forwarder shares tracking events, departed, transhipment, expected arrival, with the distributor.

Step 8: Arrival and import formalities. The distributor's agent lodges the import declaration, the destination country's customs formalities are completed, and the goods are released.

Step 9: Delivery and verification. The container is trucked to the distributor's cold store in Rotterdam, unpacked, and its temperature record is checked along with a count and inspection of the fruit. The proof of delivery is issued, stock records are updated, and payment follows the agreed terms.

What does this example demonstrate? Nine stages, one unbroken cold chain, and a handoff at every stage boundary. The goods passed from cooperative to transport operators to terminal to shipping line to the European agent to the distributor, and the information, from order to certificates to tracking events to proof of delivery, moved in step with them.

Key lesson: an export succeeds when physical handling and information are both right at every handoff. One missed cut-off or one broken temperature record can undo the whole journey.

**WORKED EXAMPLE**

### An import journey: tomato paste from Italy to Kumasi

Consider a wholesale distributor in Kumasi that supplies catering-size cartons of tomato paste to retailers across the Ashanti Region. The distributor orders a full container load from a food producer in Italy; in this example, 20 pallets holding 1,000 cartons. The agreed trade terms make the seller responsible for the carriage to Tema, while the buyer is responsible for clearing the goods on import and taking delivery in Ghana; all quantities are illustrative.

Step 1: Order and preparation. The order is confirmed, the producer verifies stock, and the cartons are counted and stacked on pallets at the Italian factory.

Step 2: Stuffing. An empty container is delivered to the factory, the 20 pallets are stuffed and secured, and the doors are sealed. The loading details are sent to the carrier in time for the documentation cut-off.

Step 3: Inland movement. The sealed container is trucked from the factory to an Italian seaport, arriving before the cargo cut-off for the booked vessel.

Step 4: Export formalities. The seller completes the export formalities of the exporting country, and the container is cleared to leave.

Step 5: Main carriage. The container is loaded and sails for Tema. During the voyage, the carrier's tracking updates show the departure, any transhipment, and a firmer expected arrival date.

Step 6: Arrival and import formalities. Before the vessel arrives, the distributor's customs representative prepares the import declaration from the seller's documents. Once the vessel berths at Tema, the declaration is processed electronically through Ghana's customs system, and customs releases the goods once it is in order and the applicable duties and taxes are settled.

Step 7: Release and collection. The released container is collected from the terminal by a truck the distributor booked in advance against the expected release date, so that no time-based storage charges accumulate.

Step 8: Inland delivery. The truck carries the container from Tema to Kumasi, where it arrives at the distributor's warehouse with its seal intact.

Step 9: Receipt and verification. Warehouse staff confirm the seal, break it, count the 1,000 cartons against the documents, and inspect for damage. The proof of delivery is signed, stock records are updated, and the invoice is settled under the agreed terms.

What does this example demonstrate? In an import journey, the buyer's heaviest responsibilities begin when the goods arrive: the formalities, the release, the collection, the

inland delivery, and the receipt are the consignee's side of the work, usually performed through representatives.

Key lesson: the same nine stages appear in every international journey. What changes between an export and an import is which party carries the load at each stage, and the trade terms agreed in the sales contract are what fix that division.

**WORKED EXAMPLE**

### Managing a delay with tracking and communication: medicines from Mumbai to Accra

Consider a pharmaceutical importer in Accra that orders 40 cartons of temperature-sensitive medicines from a supplier in Mumbai, India. The medicines are destined for hospital supply, so reliability matters as much as speed; air freight keeps the exposure time short. All details are illustrative.

Step 1: Order and preparation. The order is confirmed, the supplier picks and counts the 40 cartons, and the medicines are packed in insulated cartons prepared for temperature-sensitive air cargo.

Step 2: Handover and export formalities. The consignment is accepted by the air carrier at Mumbai against the airline's requirements, the export formalities are completed, and the shipment is booked on an air freight service to Kotoka International Airport routed through an intermediate hub.

Step 3: Carriage and a problem spotted early. The freight forwarder monitors the tracking events. Two days before the expected arrival, while reviewing the shipment documents against the destination requirements, the forwarder notices that one supporting certificate from the supplier is incomplete.

Step 4: Communication. The forwarder alerts the importer's customs representative immediately. The representative confirms that the gap would prevent release, and the supplier in Mumbai issues a corrected certificate electronically within a day. The correction is complete before the medicines land.

Step 5: Arrival and import formalities. The consignment arrives at Kotoka International Airport. The import declaration, prepared in advance, is already lodged; customs checks the documents, now complete, and releases the goods. Throughout the handling at the airport, the importer's arranged temperature-controlled storage keeps the medicines within their required range.

Step 6: Delivery and verification. The cartons are trucked to the importer's temperature-controlled warehouse in Accra, counted, and checked, including the temperature indicators on the cartons. The proof of delivery is signed, the stock records are updated, and the hospital orders are dispatched on schedule.

Now consider the same journey without communication. The certificate gap would have been discovered only after arrival, at the clearance check. The medicines would have waited at the airport while a corrected document was chased across two countries; storage charges would have accumulated; the cold chain would have faced avoidable risk; and hospital orders would have slipped.

What does this example demonstrate? Delay factors, incomplete documents, inspections, congestion, cannot always be prevented, but they can be detected early and managed. Tracking provided the visibility; communication turned that visibility into action by putting the problem in front of the one party who could fix it while there was still time.

Key lesson: the value of shipment tracking is realised through communication. Information must reach the party who can act, while action is still possible.

**WORKED EXAMPLE**

### Following the handoffs: frozen fish from Tema to Tamale

The handoff concept can be seen at its clearest in a simple domestic journey. A frozen-food company in Tema receives an order from a wholesaler in Tamale: 300 cartons of frozen fish, to be delivered within three days. There is no border crossing and no international carriage, yet the same structure of handoffs appears. All quantities are illustrative.

Step 1: Order and preparation. The order is confirmed, and 300 cartons are drawn from the cold store, counted, and stacked on pallets, remaining at freezer temperature throughout.

Step 2: Collection. A refrigerated truck arrives. The driver presents collection instructions listing 300 cartons; the cold store staff check that the trailer is at the correct temperature, load the pallets, and both parties sign the collection record. This is the first handoff: the goods pass, and a signed record of exactly what passed travels with them.

Step 3: Inland movement. The truck makes the long road journey north. The driver keeps the trailer at the set temperature, and the dispatcher follows progress through periodic updates.

Step 4: Delivery and verification. At the Tamale warehouse, the wholesaler's staff count the cartons: 298 are intact and 2 are crushed. The damage is noted on the delivery document before it is signed, while the driver is still present and the cause, loading or transit, can still be investigated.

Step 5: Records and payment. The signed proof of delivery, carrying the damage note, returns to the Tema company. Its records show the order fulfilled with two cartons damaged, the matter is resolved under the agreed terms, and the invoice is prepared to match the delivered quantity. Payment follows according to the agreed terms.

Now map the handoffs. At collection, the goods were 300 cartons and the information was the signed collection record, the very document the later invoice would rely on. At delivery, the

goods were 298 intact cartons and 2 noted as damaged, and the information was the proof of delivery carrying that damage note, on which the invoice was built rather than on the original order.

What does this example demonstrate? Even the simplest journey is a chain of handoffs, and at each one the goods, the information, and the payment records must match. Where they match, disputes are rare and quickly settled. Where they do not, the parties argue about what happened instead of running their businesses.

Key lesson: treat every handoff as a small transaction in itself. Goods pass, a record proves what passed, and those records carry the money flows that follow.

**IMPORTANT NOTE**

### Responsibility does not always sit where the goods are

A common beginner assumption is that whoever physically holds the goods is responsible for them, and that responsibility passes only when the goods physically change hands. This is not how journeys work. Responsibility for the goods, for the documents, and for the charges passes at points agreed in advance, and those points may be far from the goods' physical location.

The division of responsibility follows the trade terms of the sales contract, commonly expressed using the Incoterms 2020 rules, which allocate tasks, costs, and risk-transfer points between seller and buyer. Two consequences surprise beginners. First, under some rules the seller arranges and pays for carriage all the way to the destination while the risk of loss or damage has already passed to the buyer much earlier, for example when the goods are handed to the first carrier at origin. A buyer can therefore carry the risk of a voyage whose freight the seller is paying for. Second, the rules allocate tasks, export formalities to one side, import formalities to the other, but they do not transfer ownership of the goods, and they are not themselves the contract of carriage.

The practical habit from this lesson: at every stage, ask whose responsibility the goods are here, who is preparing the information, and who pays the charges now arising. Journeys go wrong most often at handoffs where nobody can answer those three questions with certainty.

**IMPORTANT NOTE**

### Arrival is not delivery, and tracking is not control

Two misconceptions deserve a final warning.

The first is that arrival means the job is done. Arrival begins the most delicate part of an import journey: the import formalities, the release, the deconsolidation or collection, the inland delivery, and the verification. Goods can sit for days after a vessel berths while declarations are processed, inspections are conducted, and collection transport is arranged, and once the allowed free time ends, time-based charges begin to accumulate. Delivery happens only when

the goods are verified and the proof of delivery is issued, not when the truck reaches the gate and certainly not when the ship docks.

The second is that because goods are tracked, they are under control. Tracking shows location and status; it does not clear customs, correct a wrong document, load a truck, or keep a refrigerated container at temperature. Control over a journey comes from what participants do with tracking information: correcting documents before arrival, booking trucks before release, warning customers before a delay becomes a missed promise. A party who watches a tracking screen without acting on it has observation, not control.

A successful journey is therefore measured against the whole chain, not against the arrival event: the goods arrive in the right quantity and condition, the formalities are completed in both countries, the records at every handoff are accurate, and the delivery is verified with proof of delivery and payment records aligned. The goods, the information, and the money arrive together, and that is the real end of the journey of goods from supplier to buyer.

## Practice questions (Lesson Checks)

**Practice Question 1**

A container of imported rice is discharged from a vessel at a Ghanaian seaport. The distributor in Kumasi is waiting for the goods.

Which option best describes what must happen before the distributor receives the rice?

1. The shipping line delivers the container directly to the distributor's warehouse as soon as the vessel berths.

2. The journey is complete, because arrival in the destination country ends the movement of goods.

3. Import formalities must be completed and the goods released by customs before the container can be collected and delivered inland to Kumasi. ✓

4. The consignor in the exporting country must now arrange transport from the port to the distributor's premises.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

After a shipment arrives in the importing country, the goods remain under customs control until the import formalities are completed and the goods are released. Only then can the container be collected from the terminal, or the consignment collected from a destination facility, and

moved inland to the buyer's premises. Arrival opens the destination stages of the journey; it does not end it.

Answer 1 fails because carriers do not deliver automatically at the gangway: release must be obtained and collection arranged. Answer 2 confuses arrival with completion, ignoring formalities, collection, and delivery. Answer 4 looks backwards along the chain: the consignor's responsibilities ended at the export handover, and responsibility has since passed to the importer's side.

The takeaway: between arrival at the port and delivery to the buyer stand the import formalities, the release, and the inland delivery, and each is a stage of the journey in its own right.

**Practice Question 2**

A shea butter cooperative in Tamale sends 400 cartons to a cosmetics company in Hamburg, and a freight forwarder arranges the ocean shipment on behalf of the cooperative.

Which statement best describes the consignor and the consignee of this shipment?

1. The consignor is the shipping line and the consignee is the freight forwarder, because they physically handle the goods.

2. The cooperative is the consignor, because it sends the goods, and the cosmetics company is the consignee, because it is designated to receive them. ✓

3. The cooperative is the consignee, because it prepares the goods, and the cosmetics company is the consignor, because it pays for them.

4. The freight forwarder is the consignor and the cooperative is the consignee, because the forwarder coordinates the shipment.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

The consignor is the person or organisation that sends goods to another party, and the consignee is the person or organisation designated to receive them. In this shipment the cooperative dispatches the 400 cartons from Tamale, so it is the consignor, and the Hamburg company is named to receive them, so it is the consignee. The parties who physically handle or coordinate the goods do not take either name.

Answer 1 mistakes the carrier and the forwarder for the sender and receiver; they are service providers acting on the shipment, not its endpoints. Answer 3 reverses the terms by tying them

to preparation and payment, neither of which decides the names. Answer 4 confuses coordination with sending.

The takeaway: consignor and consignee describe who sends and who is designated to receive a shipment, regardless of who transports it, coordinates it, or pays for it.

**Practice Question 3**

An importer learns from its freight forwarder, three days before the vessel arrives, that one of the documents needed for customs release is incomplete.

Which option best explains why this early communication matters?

1. Good communication makes the incomplete document acceptable to customs without correction.

2. Early warning allows the importer to correct the document before arrival, so the goods can be released promptly instead of waiting after arrival while charges accumulate. ✓

3. Tracking guarantees that the vessel will arrive on schedule, so the document problem will resolve itself.

4. Delays after arrival are always the carrier's responsibility, so the importer need take no action.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

The value of shipment tracking is realised through communication: an event or a spotted problem must reach the party who can act while action is still possible. Here, the early warning gives the importer time to obtain a corrected document before the vessel berths, so the declaration can be processed and the goods released on arrival, instead of the goods sitting in port after arrival while storage and related charges begin once the free period ends.

Answer 1 is wrong because communication supports documents but never replaces them; customs releases goods against documents that are complete and correct. Answer 3 is wrong because tracking reports progress and expected arrival but guarantees nothing, and no schedule resolves a document defect. Answer 4 is wrong because the importer's side is responsible for completing its import formalities, and delay costs money whoever is at fault.

The takeaway: early detection through tracking, followed by prompt communication to the party who can act, turns a potential crisis into a short, managed delay.

## Fact-check appendix (audit notes — not for entry)

The nine-stage chain (order and preparation, packing and stuffing, collection and inland movement, export formalities, main carriage, arrival and import formalities, deconsolidation, inland delivery, receipt and verification) is a teaching framework built from standard international trade practice — source: general business knowledge; consistent with curriculum/research/trade-transport-standards.md sections 4, 5 and 7.

Incoterms 2020 is the ICC's set of eleven rules allocating tasks, costs, and risk-transfer points between seller and buyer; the rules do not transfer ownership and are not themselves the contract of carriage — source: trade-transport-standards.md section 1.

Export clearance is the seller's task under all Incoterms 2020 rules except EXW, and import clearance is the buyer's under all except DDP; under some rules (the C rules) the seller pays carriage to destination while risk passes at shipment — source: trade-transport-standards.md section 1.

Since 1 July 2016 (SOLAS regulation VI/2) the shipper must obtain and communicate the verified gross mass of every packed container, and a container without verified gross mass may not be loaded — source: trade-transport-standards.md section 7.

Carriers set a cargo cut-off and a documentation cut-off before each sailing, and a shipment missing a cut-off usually rolls to the next departure — source: trade-transport-standards.md section 7 (carrier practice).

A phytosanitary certificate is issued by the plant-protection authority of the exporting country for plants and plant products, in IPPC model format — source: trade-transport-standards.md section 5.

Pre-arrival lodging of customs declaration data enables advance risk analysis and faster release on arrival — source: trade-transport-standards.md section 12.

Many customs administrations, including Ghana's, use a channel model sorting declarations into release without inspection, document check, or physical inspection — source: trade-transport-standards.md section 12.

Ghana's customs clearance is processed electronically through ICUMS, the national end-to-end customs management system used at the country's ports of entry and exit — source: trade-transport-standards.md section 12 (Ghana's single window operates through ICUMS) and logistics-scm-frameworks.md section 9 (Ghana anchor: end-to-end e-customs platform at all ports of entry and exit).

Carriers and terminals allow a limited period of free time after arrival, after which time-based charges (storage, demurrage, detention) accumulate; amounts and periods are purely contractual and no standard figures exist, and none are stated in this lesson — source: trade-transport-standards.md section 11.

Refrigerated (reefer) containers carry perishables, and cold-chain logistics means temperature-controlled storage and transport with continuous monitoring — source: trade-transport-standards.md section 6; logistics-scm-frameworks.md section 1.

Consolidation (groupage) combines several shippers' smaller loads in one container or vehicle, cheaper per unit but with more handling and a need for accurate marking — source: logistics-scm-frameworks.md section 6.

An ocean bill of lading records receipt of the goods and the terms of carriage; named only in passing, with document details reserved for a later lesson — source: trade-transport-standards.md section 4.

Air cargo is accepted by carriers against acceptance requirements including security screening and acceptance checks — source: trade-transport-standards.md section 8.

The order-to-receipt flow runs from order to shipment to goods receipt checked against the order, with invoicing and payment following on the basis of delivery records — source: logistics-scm-frameworks.md section 3.

Track-and-trace records shipment position and status as scan or status events from booking to delivery — source: logistics-scm-frameworks.md section 9.

All quantities, transit times, and schedules in the worked examples (18 pallets, 20 pallets, 1,000 cartons, 40 cartons, 300 cartons, a twelve-day voyage, three-day windows) are illustrative for teaching purposes only — source: writing specification rule 5.1.

# Lesson 3.2 — Modes of Transport: Sea, Air, Road, Rail, and Multimodal

## Summary

This lesson introduces the major transportation modes used in domestic and international logistics. Learners will examine sea, air, road, rail, and multimodal transportation, including their characteristics, advantages, limitations, and suitability for different types of cargo and delivery requirements.

## Learning objectives

- Identify the major modes of transportation used in logistics operations.
- Describe the characteristics of sea, air, road, and rail transportation.
- Explain the advantages and limitations of each transportation mode.
- Define multimodal transportation and explain how different modes can be combined.
- Identify factors influencing the selection of a transportation mode.
- Compare transportation modes based on cost, speed, capacity, accessibility, and cargo suitability.
- Select an appropriate transportation mode for a basic logistics scenario.

## Key terms

**Sea Freight —** The transportation of goods by vessels across seas, oceans, or other navigable waterways.

**Air Freight —** The transportation of goods by aircraft.

**Road Freight —** The transportation of goods using road vehicles such as trucks and vans.

**Rail Freight —** The transportation of goods using trains operating on railway networks.

**Multimodal Transport —** The transportation of goods using at least two different modes of transport under a single multimodal transport contract.

**Intermodal Transport —** The movement of goods using two or more transportation modes, typically in the same loading unit, without handling the goods themselves when changing modes.

**Transit Time —** The time required for goods to travel between specified points in a transportation journey.

**Freight Capacity —** The quantity or weight of cargo that a transport vehicle or service can accommodate.

**Transport Network —** A system of routes, facilities, and connections used to move goods or passengers between locations.

**Last-Mile Delivery —** The final stage of transportation in which goods are delivered from a distribution point to the intended recipient.

## Teaching content

**KEY CONCEPT**

### The major modes of transportation

The previous lesson followed goods from a supplier to a buyer and showed that a shipment passes through preparation, collection, transportation, handling, storage, and delivery. This lesson examines the transportation stage itself and asks a practical question: in how many ways can goods physically travel?

Logistics operations use five broad arrangements, called modes of transport.

1. Sea freight. The transportation of goods by vessels across seas, oceans, or other navigable waterways.

2. Air freight. The transportation of goods by aircraft.

3. Road freight. The transportation of goods using road vehicles such as trucks and vans.

4. Rail freight. The transportation of goods using trains operating on railway networks.

5. Multimodal transport. Journeys that combine two or more of these modes within one coordinated movement.

Each mode exists because it solves a different problem. A vessel can carry enormous quantities across an ocean, but it cannot deliver to a shop's door. A van can deliver to any address that a road reaches, but it cannot cross an ocean. An aircraft crosses continents in hours, at a cost that only some goods can justify. A train can haul a very heavy load over land efficiently, but only between points that a railway connects.

No single mode is at the same time the fastest, the cheapest, and the most flexible. Every transport decision is therefore a trade-off, and the practical skill is matching the mode to the goods, the distance, and the delivery requirement.

This lesson describes each mode in turn, explains how modes are combined in multimodal and intermodal movements, and closes with a framework for comparing the modes and selecting an appropriate one for a given shipment.

KEY CONCEPT

## Transit time, freight capacity, and the transport network

To compare transport modes objectively, three terms are needed. They appear throughout this lesson and in later modules, so each is defined carefully here.

1. Transit time. The time required for goods to travel between specified points in a transportation journey. Transit time is commonly quoted between the points where the carrying mode takes over and gives up the goods: airport to airport in air freight, port to port in sea freight. The total time a consignment is on the move is longer than the carrying time alone, because it includes loading, handling at terminals, waiting for scheduled departures, and, across borders, customs formalities.

2. Freight capacity. The quantity or weight of cargo that a transport vehicle or service can accommodate. Capacity differs enormously between modes. A delivery van may carry a few tonnes at most. An articulated truck carries many times more. A single large container vessel can carry thousands of containers; an earlier lesson noted that vessel capacity is commonly counted in twenty-foot equivalent units, called TEU. Capacity matters because it decides how much cargo can move in one journey, and moving more cargo in one movement usually lowers the cost per unit.

3. Transport network. A system of routes, facilities, and connections used to move goods or passengers between locations. Each mode has its own network. The road network consists of roads, depots, and delivery points. The rail network consists of railway lines, terminals, and wagons. The sea network consists of ports and shipping routes. The air network consists of airports and flight routes. A mode can serve only the places its network reaches, which is why the shape of the available networks strongly influences transport decisions in every country.

With these three terms in place, the modes can now be described and compared on a common basis.

KEY CONCEPT

## Sea freight: the largest carrier of international trade

Sea freight is the transportation of goods by vessels across seas, oceans, or other navigable waterways. It is the foundation of international trade: the greater part of the goods moving between continents travels by sea, because no other mode can match what a vessel carries in one voyage.

The defining characteristics of sea freight follow from the sheer scale of vessels.

1. Very large freight capacity. A single large container vessel carries thousands of containers, and bulk carriers and tankers move grain, ores, clinker, crude oil, and refined products in quantities measured in many thousands of tonnes.

2. The lowest cost per unit over long distance. Because a vessel spreads its operating cost across enormous cargo quantities, sea freight is normally the cheapest way to move goods between continents, unit by unit.

3. The slowest transit times. Vessels travel far more slowly than aircraft, and a voyage between continents is measured in weeks rather than days. Port handling at each end adds further time.

Sea cargo moves in two broad forms, and the distinction matters when choosing equipment.

1. Containerised cargo. Manufactured goods, packaged foods, and general merchandise are packed into standard steel containers, which container ships carry. The container protects the goods and allows easy transfer between vessel, truck, and train, a point developed later in this lesson.

2. Bulk cargo. Unpackaged homogeneous goods such as grain, ores, and clinker are poured or scooped directly into a vessel's holds and carried by dry bulk carriers. Liquids travel in tankers. Vehicles and wheeled machinery sail on specialised roll-on roll-off vessels.

An earlier lesson introduced shipping lines and their liner services, which run published routes and schedules; that is how most container cargo is booked. The next lesson examines cargo types and packaging in detail.

The main limitations of sea freight are its slow speed, its dependence on ports, and its exposure to congestion and weather along the route. A port is also never the final destination: after discharge, goods still need a road or rail journey inland. Sea freight therefore suits cargo that is heavy, bulky, or plentiful, and that is not urgent.

**KEY CONCEPT**

## Air freight: the fastest and most expensive mode

Air freight is the transportation of goods by aircraft. It is the fastest mode over long distances: where a vessel needs weeks to cross between continents, an aircraft crosses in hours, and a consignment that flies is typically delivered within a day or two of departure, airport to airport. Speed is air freight's defining advantage and the reason it exists as a freight mode.

Cargo travels in the holds of passenger aircraft and in dedicated freighter aircraft, moving through airport cargo terminals built for quick acceptance, security screening, and release. An earlier lesson named the air waybill, the transport document an airline issues for goods it carries; the airline takes the goods into its care at the origin airport and releases them at the destination airport.

The advantages of air freight follow directly from speed.

1. Shortest transit time. The decisive benefit for cargo where every day matters.

2. Less stock tied up in transit. Goods that arrive quickly start earning or selling sooner, which partly offsets the higher cost for high-value cargo.

3. Predictable schedules. Cargo is booked onto frequent scheduled flights, so the next departure is rarely far away.

The limitations are equally clear.

1. The highest cost of any mode. Air freight is charged on chargeable weight, a billing weight that reflects the space cargo occupies as well as what it weighs. Light but bulky cargo can therefore be charged as though it weighed much more. For most goods, flying costs several times more than sea freight.

2. The smallest freight capacity. An aircraft carries only a fraction of what a vessel carries, and size, weight, and safety restrictions limit what may fly at all.

3. A limited network. Air freight runs between airports equipped to handle cargo, and delivery beyond the airport needs road transport, just as sea freight does beyond the port.

Air freight therefore suits cargo that is urgent, perishable, high in value, or light. Pharmaceuticals, urgent spare parts, electronics, documents, fresh produce, and flowers are typical examples.

**KEY CONCEPT**

### Road freight: door-to-door flexibility

Road freight is the transportation of goods using road vehicles such as trucks and vans. It is the mode most people see every day, and the only mode that, in normal practice, completes a delivery to an ordinary address. It dominates regional and local distribution and carries the first and last legs of most international journeys, whatever mode crosses the ocean or continent in between.

The defining characteristics of road freight are flexibility and reach.

1. Door-to-door delivery. A truck can collect from a farm, factory, or warehouse and deliver to a shop, a site, or a customer's gate without transhipment. No other mode offers this.

2. A vehicle for every load. Road freight uses everything from small vans to large articulated trucks, so the vehicle can be matched to the size of the consignment.

3. Flexibility in timing and routing. Road transport can depart when it is needed and adjust its route, which fixed networks and published schedules do not allow in the same way.

Road freight also owns the final stage of nearly every supply chain. Last-mile delivery is the final stage of transportation in which goods are delivered from a distribution point to the intended recipient. It is the stage customers actually see, and it is almost always performed by road.

The advantages of road freight are door-to-door service, speed over short and medium distances, and wide availability, since roads reach places that ports, airports, and railway lines do not.

The limitations are equally practical.

1. Limited freight capacity per vehicle. A truck carries a tiny fraction of what a train or vessel carries, so cost per unit rises when large volumes move long distances by road alone.

2. Higher cost per unit over long hauls. Where rail or sea can take over the long part of a journey, they usually do it more cheaply.

3. Exposure to road conditions. Traffic congestion, road quality, weather, and border formalities all affect road transit times.

Road freight therefore suits regional distribution, collection and delivery legs, and any movement that must end at a specific address.

**KEY CONCEPT**

### Rail freight: heavy loads over land

Rail freight is the transportation of goods using trains operating on railway networks. Its defining strength is efficiency over land: one train, hauled by one or a few locomotives, can move a load that would otherwise require a very large number of trucks. Where a suitable network exists, rail is therefore an economical way to carry heavy or bulky cargo long distances between cities and industrial areas.

The characteristics of rail freight follow from the nature of the network.

1. Large freight capacity per movement. Trains carry bulk commodities in specialised wagons and containers on flat wagons, combining many wagonloads into one movement.

2. Fixed routes and terminals. A train runs only where track exists, between terminals equipped to load and unload it. This makes rail services steady and repeatable where they operate.

3. Efficiency for heavy, dense cargo. The cost per tonne over land is low for cargo that fills the train's capacity, which is why mining and agricultural regions use rail heavily.

The advantages of rail freight are its economy for heavy bulk over long land distances, its large capacity per movement, and its independence from road congestion.

The limitations decide where rail is used.

1. The network limits reach. Rail can serve only points joined by track. Where lines do not exist, or do not connect the right places, rail is simply not an option, whatever its advantages on paper. How much railway infrastructure exists differs from one country and region to another.

2. Rail is rarely door-to-door. Goods almost always travel the first and last parts of the journey by road, moving between the origin, the rail terminal, and the final destination. Rail performs the middle, long-haul leg.

3. Dependence on schedules and terminals. Cargo must fit the service's timetable and be handed over at a terminal, which adds handling compared with a single truck journey.

Rail freight therefore suits heavy, non-urgent bulk cargo such as minerals, grain, timber, and clinker, and containerised goods moving long distances over land between connected terminals.

**KEY CONCEPT**

### Multimodal and intermodal transport

Sea, air, road, and rail each cover a different part of a journey, so long movements normally use more than one of them. Two terms describe how modes are combined, and they are related but not identical.

Multimodal transport is the transportation of goods using at least two different modes of transport under a single multimodal transport contract. One operator, called the multimodal transport operator, takes responsibility for the whole journey from origin to destination, even where different carriers, such as a shipping line and a trucking company, each perform one leg. The shipper deals with one contract, one responsible partner, and one point of contact.

Intermodal transport is the movement of goods using two or more transportation modes, typically in the same loading unit, without handling the goods themselves when changing modes. The loading unit is usually a container: goods are packed once at origin, and at every change of mode the container itself is lifted from ship to train to truck, while the goods inside are never touched.

The container is what makes such movements practical, because it is built to a standard that ships, trains, and trucks can all carry. Special types extend the principle: refrigerated containers carry perishables, open-top and flat-rack containers carry tall or heavy items, and tank containers carry liquids.

Packing goods once and lifting the sealed unit at each change of mode brings real benefits: less handling means less damage and theft, quicker transfers between modes, and simpler security because the container stays sealed.

The difference between the two terms is a common source of confusion, so state it precisely. Intermodal transport describes how the goods physically travel: in one and the same loading unit, untouched at each change of mode. Multimodal transport describes how the journey is contracted and who answers for it: one contract and one operator responsible end to end. The two are not competing choices. A single door-to-door container shipment can be intermodal in its physical form and multimodal in its contract at the same time.

KEY CONCEPT

## How modes combine in one journey

The modes studied so far rarely appear alone in international trade. A typical import or export is a sequence of legs, each performed by the mode that suits it best; the previous lesson's journey from supplier to buyer usually crosses modes along the way.

A containerised import shows the pattern.

1. Road leg at origin. A truck collects the sealed container from the overseas supplier's premises and delivers it to the port of loading.

2. Sea leg. The container ship carries the container across the ocean to the destination port, such as Tema Port for a Ghanaian import.

3. Road leg at destination. After arrival formalities, a truck carries the container inland to a warehouse in Kumasi or another city.

The same logic appears in exports by air: a farm's produce travels by road to Kotoka International Airport, flies to a European airport, and travels by road again to the buyer's distribution centre. The aircraft does the long, fast leg; trucks collect and deliver at each end.

Notice what each mode contributes. The vessel or aircraft covers the long international leg, where its capacity or speed is worth the cost. Road transport covers the short legs, where door-to-door flexibility matters more than capacity. Where railways connect the right places, a train can take over an inland leg instead of trucks.

Who coordinates these legs depends on the contract. Under a multimodal transport contract, as defined earlier, one operator is responsible for arranging the whole chain, and the shipper holds that operator accountable from door to door. The same physical journey can also be assembled from separate contracts, one per leg, which the shipper or its freight forwarder then coordinates.

This is why mode selection is rarely a choice of one mode only. The real question is usually which combination of modes, contracted how, moves the goods best.

KEY CONCEPT

## Factors that influence the choice of mode

Choosing a transport mode is a balancing exercise. The main factors, and the questions they raise, are the following.

1. Nature of the cargo. Perishable goods need short transit times and continuous temperature control. High-value goods justify spending more on speed and security. Fragile goods favour fewer handlings. Heavy machinery and bulk commodities favour modes with large capacity. Dangerous goods face additional restrictions that differ by mode.

2. Urgency and required transit time. If the goods must arrive by a date, the shortlisted modes must be able to achieve it. Non-urgent goods can travel more slowly and more cheaply.

3. Cost of transport. Freight cost is compared against the value of the goods and the budget for the order. The differences between modes are large: over long distances, sea freight costs a fraction of air freight for the same weight. Paying for speed the goods do not need wastes money.

4. Quantity to be moved. The freight capacity each mode offers must match the consignment. A full container or truckload can be booked directly, while smaller consignments can share space through consolidation, a service an earlier lesson described under freight forwarders.

5. Accessibility and the transport network. The chosen mode must actually run between the relevant places; inland origins and destinations depend on road or rail connections for the first and last legs.

6. Reliability. A mode and route are judged not only on average speed but on how predictably they meet schedules, because downstream plans, from production runs to shop shelves, depend on promised arrival dates.

7. Total cost, not freight cost alone. Faster modes cost more to move goods but reduce the time stock is tied up in transit and the risk of running out; slower, cheaper modes do the reverse. A considered choice weighs the freight bill against these wider effects.

No single factor decides the answer. The appropriate mode best balances these factors for the specific shipment, as the worked examples later in this lesson demonstrate.

**KEY CONCEPT**

## A framework for comparing the modes

With each mode described and the selection factors listed, the modes can now be compared directly. Five lenses cover the comparison: speed, cost, capacity, accessibility, and cargo suitability. The statements below are directional comparisons for building a shortlist; exact figures vary by route, season, and service.

1. Speed. Air freight is the fastest over long distances. Sea freight is the slowest, with voyages measured in weeks. Road freight is fast and direct over short and medium distances. Rail moves steadily over land where track exists. A multimodal journey's total transit time is the sum of its legs plus the handling at each change of mode.

2. Cost. Per unit of cargo over long distance, sea freight is normally the cheapest. Rail is economical for heavy loads over land. Road is efficient for short distances but comparatively expensive over long hauls. Air freight is the most expensive by a wide margin.

3. Capacity. Sea freight offers the greatest freight capacity, a single vessel carrying what would take thousands of trucks. Rail offers large capacity per train on its network. Road is limited per vehicle. Air offers the smallest capacity of all.

4. Accessibility. Road reaches nearly every address, making it the most accessible mode and the natural carrier of the last mile. Sea serves only ports, air only cargo airports, and rail only its own network. Multimodal combinations extend reach by connecting these networks.

5. Cargo suitability. Urgent, perishable, high-value, and light cargo suits air freight. Heavy, bulky, or plentiful non-urgent cargo suits sea freight. Heavy bulk over land between connected points suits rail. Regional distribution and every delivery to a specific address suit road.

Read together, the five lenses show why the modes coexist: each is the best answer to a different question. The worked examples that follow apply the framework to three Ghana-linked consignments and one international route.

**WORKED EXAMPLE**

### Flying fresh green beans from Accra to Europe

A commercial farm near Accra grows vegetables for export. A supermarket buyer in Europe places an order for fresh green beans, to arrive within days of harvest so the beans can reach the shelves while still fresh. This week the order is 150 cartons, and all quantities in this example are illustrative for teaching purposes.

Step 1: The requirement is examined. Green beans have a shelf life of only a few days, and the buyer pays a premium for freshness. Sea freight from Tema Port would take weeks, longer than the beans can stay fresh, so sea freight is ruled out however cheap it is. Air freight is selected: the cargo's perishability demands speed, and its high value per unit justifies the higher freight cost.

Step 2: Harvest and the cold chain. The beans are harvested, sorted, and packed into cartons, then moved into a cold store to remove field heat. Temperature control continues at every stage that follows, because a break in the cold chain would shorten shelf life just as effectively as a slow mode would.

Step 3: Road leg to the airport. A refrigerated truck collects the 150 cartons from the farm and delivers them to the air cargo terminal at Kotoka International Airport. The road leg is short but essential: no aircraft will collect cartons from a farm.

Step 4: Acceptance and documentation. At the terminal, the shipment is checked in, screened, and documented. The airline or its agent issues an air waybill, recording the goods received for carriage to the destination airport.

Step 5: The air leg. The cartons fly to the destination airport in Europe within hours.

Step 6: Arrival and delivery. The buyer's agent completes the arrival formalities, and a truck delivers the beans to the buyer's distribution centre a day or two after packing.

What does this example demonstrate? The mode was chosen from the cargo's requirements outward: perishability demanded speed, and value justified cost. Road freight performed the first and last legs, so the shipment was already a combination of modes in practice.

Key lesson: air freight is not chosen despite its cost. It is chosen when the goods' shelf life and value make speed worth more than the saving a slower mode would offer.

**WORKED EXAMPLE**

### Importing rice by sea through Tema Port

An importing and distribution company in Accra buys packaged rice from a supplier in Asia for the Ghanaian market. The order is for 400 tonnes of rice packed into twenty containers. All figures in this example are illustrative for teaching purposes.

Step 1: The requirement is examined. Rice is non-perishable with a long shelf life, the volume is large, and the company plans its purchases months in advance. Speed adds nothing the goods do not need, while cost per tonne matters greatly at this quantity. Sea freight is selected: for a heavy, non-urgent consignment between continents it offers the lowest cost per unit, and air freight would cost many times more for no benefit the goods can use.

Step 2: Booking and loading. The company works through a freight forwarder, which books space for the containers with a shipping line, as an earlier lesson described. At the port of loading, the rice is packed into the containers, which are sealed and delivered to the terminal in time for the sailing.

Step 3: The sea leg. The containers travel on a container ship to Tema Port. The voyage is measured in weeks, the longest transit time of any mode, and the company's purchasing plan was built around it.

Step 4: Arrival and the inland leg. After the arrival formalities at the port, trucks collect the containers and carry them to the company's warehouse in Accra, where the cartons are unloaded. The same sealed container did the sea leg and the road leg, and the goods themselves were not handled between the two.

Step 5: The comparison made explicit. Sea freight took the longest time of any mode but delivered 400 tonnes at the lowest cost per tonne. The rice's long shelf life made the weeks at sea harmless, and the saving helped the imported staple reach the market at a competitive price.

What does this example demonstrate? When cargo is heavy, bulky, or plentiful and not urgent, sea freight's capacity and low cost per unit outweigh its slow transit time.

Key lesson: the slowest and cheapest mode is often the correct choice, because transit time is only a defect when the goods need speed.

**WORKED EXAMPLE**

### Distributing cement by road in Ghana

A building-materials distributor in Takoradi supplies bagged cement to hardware shops and construction sites in Takoradi, Cape Coast, and the surrounding towns. Cement arrives at the distributor's depot in bulk loads and leaves it in bags. This example follows the outbound distribution.

Step 1: Inbound supply. Trucks deliver cement from the production plant to the distributor's depot, where it is stored and recorded.

Step 2: Orders arrive. Hardware shops order anything from a few dozen bags to a part-load; construction sites order scheduled quantities tied to their building programmes.

Step 3: Matching vehicles to loads. The distributor loads each order onto a suitable road vehicle: large deliveries onto an articulated truck that may serve several customers on one route, smaller orders onto a light truck or van.

Step 4: Door-to-door delivery. Each delivery ends at a specific address: a shopfront in a trading street or a site gate. Drivers offload, the receiving party checks and signs for the bags, and the paperwork confirms delivery.

Step 5: The last mile. Every one of these journeys ends with the final stage of transportation in which goods are delivered from a distribution point to the intended recipient. The distributor's vehicles perform this stage many times a day.

Why not another mode? Sea freight cannot reach any of these addresses; it serves ports. Rail freight would help only if track connected the depot to each customer, which address-by-address delivery rules out by its nature. Air freight has no role at these distances. Road freight is the only mode that can complete these movements, and its flexibility in vehicle size, timing, and routing suits orders that differ every day.

What does this example demonstrate? Regional distribution is road freight's natural territory. The mode's door-to-door reach, its range of vehicle sizes, and its freedom from fixed schedules match a daily pattern of many small, different deliveries.

Key lesson: the question is rarely which mode is strongest in the abstract. It is which mode can complete the movement as the goods actually require, and for delivery to an ordinary address, only one mode can.

**WORKED EXAMPLE**

### One order from China to Europe, three possible modes

A manufacturer in Shenzhen, in China, supplies a retailer in Hamburg, in Germany. The retailer has confirmed a large order of consumer electronics and needs a delivery plan for the season. Three transport options are considered, and more than one may be used. The comparison is qualitative and quotes no figures.

Step 1: Sea freight, the default for volume. Containers leave a Chinese port by sea, pass along maritime shipping routes to a European port, and continue inland by road or rail. Sea freight offers the lowest cost per unit by a wide margin and the greatest capacity, but the longest transit times, measured in weeks. It suits the main volume of the order, which is forecast well in advance.

Step 2: Rail freight, the middle option over land. Overland rail services connect China with cities in Europe, carrying containers across the continent. Rail's position sits between the other two: faster than sea, slower than air, and priced between them. It suits goods too time-sensitive to spend weeks at sea but not urgent enough to justify the cost of flying, such as replenishment stock during the season.

Step 3: Air freight, for what cannot wait. The launch quantity and any urgently needed items fly from a Chinese airport to a European airport within days, at the highest cost per unit. The retailer accepts this cost on a small part of the order because launch dates carry sales it does not want to miss.

Step 4: Combining the options. The retailer and the manufacturer split the order: launch stock by air, mid-season replenishment by rail, and the bulk by sea. Each part travels by the mode whose profile fits its timing, and every journey still uses road legs at each end, in the combination patterns described earlier in this lesson.

What does this example demonstrate? The same goods, on the same route, for the same buyer can sensibly travel three different ways, because each mode answers a different need: cost for volume, balance for replenishment, and speed for urgency.

Lessons from the example: mode selection is a portfolio decision as much as a per-shipment decision, and the speed and cost profiles of sea, rail, and air give a trader a ladder on which to place each part of its demand.

**IMPORTANT NOTE**

### There is no single best mode of transport

Beginners often ask which mode of transport is best, expecting a single answer. There is none. The comparison framework in this lesson has one purpose: to replace the search for a best mode with the search for the appropriate mode for this cargo, this route, and this delivery requirement.

Two errors show why. Calling air freight the best mode ignores that its cost is wasted on cargo that is not urgent: flying 400 tonnes of rice would multiply the freight bill without helping the goods. Calling sea freight the best mode ignores that its weeks of transit destroy value for anything perishable: the cheapest quote is worthless if the goods arrive spoiled. The same test applies to road and rail; each is the worst choice for someone else's cargo.

A related error is judging a decision on the freight bill alone. The cheapest transport can be the most expensive choice overall if it delivers late, forces the business to hold extra stock to cover its slowness, or exposes fragile goods to unnecessary handling. Total cost, including these wider effects, is what a business actually pays.

Three questions sort most decisions. What are the goods, and what do they tolerate? When must they arrive, and how certain must that date be? What does each available option cost, in money and in consequences? A mode chosen from the answers will rarely be glamorous, but it will be defensible.

Remember: in transport, best is not a property of the mode. It is a fit between the mode and the shipment.

**IMPORTANT NOTE**

### Multimodal is not another word for intermodal

The two terms for combined transport are often used interchangeably, and the confusion runs in one particular direction: people say intermodal when they mean the single-contract arrangement. The distinction is worth fixing, because the two words answer different questions.

Multimodal transport is about the contract and responsibility. It means carriage by at least two modes under a single multimodal transport contract, with one operator responsible for the goods from origin to destination. If you know whom to turn to when something goes wrong anywhere along the journey, the arrangement is multimodal in character.

Intermodal transport is about the physical unit. It means the goods travel in the same loading unit, typically a container, across two or more modes without the goods themselves being handled at the changes of mode. Under an intermodal movement the contracts are commonly arranged separately per leg, so there may be several responsible carriers rather than one.

Both features can appear in one shipment. A container packed at a factory, lifted unchanged from truck to ship to train, and delivered to a warehouse under one contract with one responsible operator is intermodal in its physical form and multimodal in its contract at the same time. Neither feature implies the other: goods can change mode in the same container under separate contracts, and a single contract can cover cargo that is handled and re-packed between legs.

When you meet the terms, ask two questions in order. Who is responsible, and under how many contracts? And are the goods themselves handled when the mode changes? The answers, not the labels, tell you which arrangement you have.

**IMPORTANT NOTE**

### A mode must exist on the route before it can be chosen

The comparison framework ranks modes as though all of them were always available. In practice, availability decides before preference does. A mode can only be selected where its network actually connects the origin and the destination, and every mode's network has edges.

Sea freight requires a port at the loading end and a port at the receiving end. Air freight requires airports equipped to handle cargo, which not every airport is. Rail freight requires track between the right places; where a country's or region's railway network is thin, rail drops out of the shortlist regardless of its advantages on paper. Road freight needs a usable road, and even road has limits where terrain or road condition makes routes unreliable.

Inland geography matters as well. A business located far from any port or cargo airport depends on road or rail connections to reach the long-distance modes at all, and the final stage of almost every journey, the last-mile delivery, is completed by road. This is why road freight appears in nearly every international movement even when it crosses no border itself.

The practical habit this teaches is to map before costing. Before comparing quotes, establish how the origin connects to the destination: which ports, which airports, which railway lines, which roads. The options that do not physically connect the two points are not cheaper alternatives; they are not alternatives at all.

Because infrastructure differs from country to country and changes over time as networks grow, the shortlist of available modes should be re-checked for each route rather than assumed from experience elsewhere.

## Practice questions (Lesson Checks)

**Practice Question 1**

A fruit and vegetable exporter near Accra receives an order from a supermarket buyer in Europe for fresh green beans. The beans remain fresh for only a few days after harvest, and the buyer pays a premium price for fresh produce.

Which option best explains why air freight is the appropriate mode for this consignment?

1. Air freight is the cheapest mode of transport for goods moving between continents.

2. The beans' short shelf life requires the fastest available transit, and their high value per unit justifies paying the higher cost of air freight. ✓

3. Air freight carries goods in sealed containers that are not opened until they reach the buyer.

4. Air freight removes the need for temperature-controlled handling before and after the flight.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Air freight's defining advantage is the shortest transit time of any mode, and its defining limitation is the highest cost. For fresh green beans, both facts point the same way: a shelf life of a few days rules out the weeks that sea freight requires, and the produce's premium value per unit makes the air freight bill worth paying. This is exactly the profile of cargo air freight exists to carry: urgent, perishable, high-value, and light.

Answer 1 fails because it reverses the cost relationship: sea freight, not air freight, is the cheapest mode over long distances. Answer 3 confuses air freight with containerised sea transport; air cargo travels on aircraft pallets and in airline unit load devices, not in sealed shipping containers. Answer 4 fails because perishable goods need an unbroken cold chain regardless of mode, including pre-cooling and refrigerated transport before the flight.

The takeaway: a mode is appropriate when its profile fits the cargo's requirements, and for perishable high-value produce, speed is worth more than the cost saving of a slower mode.

**Practice Question 2**

An importer in Kumasi buys goods from a supplier in Asia. The goods travel by sea to Tema Port and then by road to Kumasi. The importer appointed one company under a single contract that covers the whole journey, and that company is responsible for the goods from the supplier's door to the warehouse door in Kumasi, even though a shipping line performs the sea leg and a trucking company performs the road leg.

Which statement best describes this arrangement?

1. It is intermodal transport, because a single contract covers carriage by more than one mode.

2. It is multimodal transport, because at least two modes are used under a single contract with one operator responsible for the entire journey. ✓

3. It is sea freight only, because the vessel carries the goods for the greater part of the distance.

4. It is road freight only, because the delivery into Kumasi is performed by a truck.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Multimodal transport is the transportation of goods using at least two different modes of transport under a single multimodal transport contract, with one operator responsible for the goods from origin to destination. The shipment described uses sea and road, is covered by one contract, and names one responsible company, which satisfies every element of the definition.

Answer 1 fails because it swaps the definitions, which is the most common confusion in this topic: the single contract and the single responsible operator are the marks of multimodal transport, while intermodal transport describes goods travelling in the same loading unit across modes without being handled, often under a separate contract for each leg. Answers 3 and 4 fail because naming a single mode contradicts the arrangement: the journey deliberately combines two modes, and neither the longer leg nor the final leg alone describes it.

The takeaway: when two or more modes move under one contract with one responsible operator, the arrangement is multimodal, whatever the modes are and whichever carries the goods furthest.

**Practice Question 3**

A food company must move 200 tonnes of dried maize from a depot to its factory in another region. A railway line connects the two towns, and both have goods terminals. The maize is non-perishable, and delivery is planned several weeks in advance.

Which option best explains why rail freight is a suitable choice for this movement?

1. Rail freight delivers door-to-door from the depot to the factory without any road transport.

2. Rail freight is the fastest of all transport modes available for this cargo.

3. Rail freight can move heavy, non-urgent bulk cargo efficiently over land between points connected by the rail network, with road vehicles covering the short legs at each end. ✓

4. Rail freight is available only for consignments that are too small for sea freight to accept.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Rail freight's defining strength is efficiency over land for heavy, dense, non-urgent cargo, and the scenario supplies every condition it needs: a large tonnage, no urgency, and a network that connects the two points. Because rail runs between terminals rather than between doors, the first and last legs of the journey are normally completed by road, which is exactly the combination pattern this lesson describes.

Answer 1 fails because rail is rarely door-to-door; a train serves its own network and terminals, not a depot gate or a factory yard, so road legs at each end are almost always required. Answer 2 fails because air freight, not rail, is the fastest mode, and speed is not rail's advantage in any case; rail's advantage is capacity and cost efficiency for bulk. Answer 4 fails because rail is not a fallback for consignments other modes reject; it is a first-choice mode for heavy bulk where the network allows.

The takeaway: where a network connects the right places, rail is often the most economical way to move heavy, non-urgent bulk over land, supported by road at both ends.

### Fact-check appendix (audit notes — not for entry)

Definitions of sea freight, air freight, road freight, rail freight, multimodal transport, intermodal transport, transit time, freight capacity, transport network, and last-mile delivery follow the curriculum key terms for this lesson. — source: curriculum/lesson_briefs/module-03/lesson-3.2.md (Key Terms).

Multimodal transport is carriage by at least two modes on the basis of a single multimodal transport contract, with one multimodal transport operator responsible end to end even where sub-carriers perform individual legs. — source: curriculum/research/logistics-scm-frameworks.md section 6 (multimodal transport).

Intermodal transport means goods travel in one and the same loading unit (container, swap body) across two or more modes without handling the goods; contracts are often separate per leg, and the multimodal/intermodal distinction is taught as common usage, not as hard law. — source: curriculum/research/logistics-scm-frameworks.md section 6 (intermodal transport) and its DO NOT CLAIM list.

The principal sea cargo vessel types are container ships, dry bulk carriers (grain, ores, clinker), tankers, and roll-on roll-off vessels. — source: curriculum/research/trade-transport-standards.md section 7 (principal vessel types).

A single large container vessel can carry thousands of containers, and vessel capacity is commonly counted in twenty-foot equivalent units (TEU). — source: curriculum/research/trade-transport-standards.md section 6 (TEU/FEU), used qualitatively.

Container types include refrigerated (reefer) containers for perishables, open-top and flat-rack containers for over-height or heavy items, and tank containers for liquids. — source: curriculum/research/trade-transport-standards.md section 6 (special types).

Container sizing in the rice example is consistent with the research sheet's guidance that a standard 20-foot container carries roughly 28 tonnes and that actual loads are usually capped by road weight limits; figures stated in the example are illustrative only. — source: curriculum/research/trade-transport-standards.md section 6 (safe phrasings).

Air cargo is charged on chargeable weight, a billing weight reflecting the space cargo occupies as well as its actual weight (the IATA volumetric principle). — source: curriculum/research/trade-transport-standards.md section 8 (chargeable weight); taught qualitatively, formula deliberately omitted at beginner level.

Air cargo travels in passenger aircraft holds and freighter aircraft, loaded on aircraft pallets and in airline unit load devices rather than shipping containers. — source: general business knowledge.

Cold-chain logistics is temperature-controlled storage and transport for perishables with continuous temperature control. — source: curriculum/research/logistics-scm-frameworks.md section 1 (cold-chain logistics).

Total-cost trade-off: slower, cheaper modes raise in-transit and cycle inventory costs; faster modes cost more per unit but reduce inventory holding and stockouts. — source: curriculum/research/logistics-scm-frameworks.md section 6 (cost trade-off).

Directional mode comparisons used throughout (sea cheapest per unit and slowest over long distance; air fastest and most expensive; rail economical for heavy bulk over land; road most accessible and the carrier of the last mile; sea the largest capacity per movement; most intercontinental goods trade moving by sea) are qualitative orderings of general business knowledge; no figures are quoted. — source: general business knowledge (directional only).

Overland rail services connect China with cities in Europe, positioned qualitatively between sea and air on speed and cost; no transit times, rates, or route specifics are claimed. — source: general business knowledge (qualitative only).

Tema Port and Kotoka International Airport are used as illustrative locations only; no claim is made about the capacity, services, or performance of any Ghanaian infrastructure, because no Ghana research sheet was available for this draft. — source: general geography; locations sanctioned by the writing specification section 4.

The green bean, rice, and cement worked examples are illustrative teaching scenarios; no claim is made about actual Ghanaian trade volumes, companies, or contracts. — source: illustrative example per writing specification section 4.

# Lesson 3.3 — Cargo Types, Packaging, and Handling

## Summary

This lesson introduces the different types of cargo transported through logistics networks and the methods used to package, protect, and handle them. Learners will examine general cargo, bulk cargo, containerised goods, perishable products, fragile goods, and dangerous goods, while understanding how cargo characteristics influence packaging, handling, storage, and transportation requirements.

## Learning objectives

- Define cargo and identify common cargo classifications.
- Distinguish between general cargo, bulk cargo, and containerised cargo.
- Identify cargo categories requiring special handling or storage conditions.
- Explain the purpose of packaging in protecting goods during transportation.
- Describe common packaging materials and cargo-handling equipment.
- Recognise the importance of cargo markings, labels, and handling instructions.
- Explain how improper packaging and handling can result in cargo damage or loss.
- Select suitable basic packaging and handling methods for different cargo types.

## Key terms

**Cargo —** Goods transported by ship, aircraft, train, truck, or another means of transportation.

**General Cargo —** Goods transported as individual packages, units, or consignments rather than as unpackaged bulk commodities.

**Bulk Cargo —** Large quantities of unpackaged commodities transported directly in a vessel's hold, tank, or other suitable transport compartment.

**Containerised Cargo —** Goods packed and transported inside standardised freight containers.

**Breakbulk Cargo —** Cargo transported as separate packages or units rather than in containers or as unpackaged bulk.

**Perishable Cargo —** Goods that may deteriorate or spoil unless transported and stored under suitable conditions.

**Fragile Cargo —** Goods that are susceptible to damage from impact, vibration, pressure, or improper handling.

**Dangerous Goods —** Substances or articles that present hazards during transportation and are subject to applicable safety regulations.

**Packaging —** Materials and methods used to contain, protect, preserve, and prepare goods for handling, storage, and transportation.

**Cargo Handling —** The activities involved in loading, unloading, lifting, moving, stacking, and securing goods.

**Palletisation —** The arrangement and securing of goods on pallets to facilitate handling, storage, and transportation.

**Cargo Marking —** The application of identification details, handling instructions, or other relevant information to cargo packages.

## Teaching content

**KEY CONCEPT**

### What cargo is and how it is classified

Walk through any market, warehouse, or construction site in Ghana and you are surrounded by cargo waiting to move: bags of cement at a distributor, drums of shea butter ready for export, sacks of cocoa at a buying centre. In logistics, all of these are cargo. Cargo means goods transported by ship, aircraft, train, truck, or another means of transportation. The previous lesson introduced the transport modes that move goods; this lesson looks at the goods themselves, because the nature of the cargo decides almost everything about how it must be packaged, handled, stored, and carried.

Cargo is classified so that everyone in the chain can make sensible operational decisions. The classification is practical, not academic: it determines what equipment lifts the goods, what

packaging protects them, what documents describe them, and what the movement costs. The main classification questions used in this lesson are the following.

1. How does the cargo travel? Goods that move as individual packages or units are general cargo; goods that move unpackaged in very large quantities are bulk cargo.

2. Is the cargo packed into a container? Goods packed inside standardised freight containers are containerised cargo, whatever the goods themselves may be.

3. Does the cargo have special characteristics? Some goods deteriorate quickly (perishable cargo), some break easily (fragile cargo), and some present hazards (dangerous goods).

4. Is the cargo unusually large or heavy? Exceptional items belong to project or oversized cargo.

The same goods can fall into different classifications depending on how they are packed and presented for transport, a point this lesson returns to more than once. What never changes is the principle: identify the cargo type first, and the packaging, handling, storage, and transport decisions follow from it.

**KEY CONCEPT**

## General and breakbulk cargo

General cargo means goods transported as individual packages, units, or consignments rather than as unpackaged bulk commodities. It is the category most people picture first, because most goods in everyday trade travel this way: cartons, sacks, drums, crates, bales, and pieces of equipment handled one unit at a time.

General cargo has three practical features.

1. It moves in countable units. A consignment of 500 cartons can be counted, marked, and checked piece by piece, making receipt and delivery verification straightforward.

2. Each unit carries its own protection and identification. The carton, sack, drum, or crate is itself the packaging that protects the goods, and each package can carry marks and labels, subjects this lesson takes up shortly.

3. It needs more handling than bulk. Every unit must be lifted, moved, stacked, and unstacked individually or on pallets, so handling effort is greater than for loose commodities.

Cement illustrates the familiar Ghanaian case: it leaves the factory in bags and moves to distributors and construction sites as bagged general cargo, counted sack by sack, even though the product inside is a powder that could in principle be moved loose.

The term breakbulk cargo is closely related. Breakbulk means cargo transported as separate packages or units rather than in containers or as unpackaged bulk. Historically, breakbulk described goods loaded piece by piece into the general holds of conventional ships; the term is

still used for separate-piece cargo where the pieces are too large or awkward for containers, or where the ports involved are not fully containerised. Breakbulk is therefore part of the general cargo family: all breakbulk is general cargo in separate units, though modern general cargo is very often palletised or containerised as well.

**KEY CONCEPT**

### Bulk cargo: dry and liquid

Bulk cargo means large quantities of unpackaged commodities transported directly in a vessel's hold, tank, or other suitable transport compartment. Nothing separates the goods from the carrying compartment: the cargo is poured, pumped, or dropped straight in. Bulk cargo divides into two families.

1. Dry bulk. Free-flowing solid commodities poured into holds or silos: grains, ores, sand, gravel, fertiliser, sugar, and clinker, the raw material ground into cement. Specialised vessels with large open holds carry dry bulk, and silos and sheds store it ashore. Handling uses grabs, conveyor belts, and loaders rather than forklifts, because there are no packages to pick up.

2. Liquid bulk. Unpackaged liquids pumped directly into tanks: crude oil, refined fuels, chemicals, and vegetable oils. Tankers carry liquid bulk by sea, pipelines and hoses move it ashore, and storage tanks hold it until it is needed. There is no packaging in the ordinary sense; the tank is the compartment.

Ghana's trade includes both families, and one national example deserves attention: cocoa beans, one of Ghana's best-known agricultural exports, move toward the ports in enormous volumes. When the beans travel loose in large quantities they are handled as a dry agricultural bulk commodity; when the same beans are packed in jute sacks, each sack becomes a unit of general cargo. The product has not changed, only the form in which it travels, and the classification changes with it.

Bulk handling is efficient because it removes packaging: vast quantities move quickly through the right terminals, but the goods are exposed, so quality depends on clean compartments, dry storage, and careful handling at every transfer point.

**KEY CONCEPT**

### Containerised cargo and the standard container

Containerised cargo means goods packed and transported inside standardised freight containers. The freight container is a standard steel box, strong enough to be lifted and stacked, sized the same way in every country, and transferable between ship, truck, and rail without handling the goods inside; it is what makes the multimodal movement introduced in the previous lesson practical.

Containers are built to international standards that fix their dimensions and strength. The commonly quoted figures for the two most common dry containers follow.

The twenty-foot container measures approximately 5.9 metres long, 2.35 metres wide, and 2.39 metres high inside, giving roughly 33 cubic metres of space. It may be loaded to a maximum gross mass of 30,480 kilograms, the combined weight of the empty box and its contents, of which approximately 28,250 kilograms is available as payload for cargo.

The forty-foot container measures approximately 12 metres long with the same width and internal height, giving roughly 67.5 cubic metres, yet its payload, approximately 26,700 kilograms, is actually lower than a twenty-foot box. A useful rule follows: heavy, dense cargo goes in twenty-foot containers, while light, bulky cargo suits forty-foot boxes. The high-cube forty-foot container adds internal height, approximately 2.7 metres, for cargo that stacks high but weighs little.

Three weight terms describe any container. Tare is the weight of the empty container. Payload is the maximum weight of cargo it may carry. Maximum gross is the permitted total of the two.

Capacity is counted in standard units: the TEU, twenty-foot equivalent unit, counts one twenty-foot container, and the FEU, forty-foot equivalent unit, counts one forty-foot container. One FEU equals two TEU, and a high cube still counts as one FEU.

Beyond the dry box, standard container types answer special needs: the refrigerated container, or reefer, carries perishables under temperature control; the open top, with a removable tarpaulin roof, is loaded from above for tall or long cargo; the flat rack, with no sides or roof, carries machinery and other out-of-gauge items; and the tank container, a tank built into a twenty-foot frame, carries liquids, gases, and powders, often dangerous goods.

**KEY CONCEPT**

### Cargo that needs special care: perishable and fragile goods

Some cargo survives an ordinary journey without complaint; some does not. Two categories in particular impose conditions on packaging, handling, and storage.

Perishable cargo means goods that may deteriorate or spoil unless transported and stored under suitable conditions. Frozen tuna landed for processing and export, fresh fruit and vegetables, dairy products, cut flowers, and many medicines are perishable. What they need is the cold chain: an unbroken sequence of temperature-controlled storage and transport, from the chilled store at origin, through the refrigerated container or temperature-controlled truck, to the cold room at destination, with the temperature monitored throughout. The chain is only as strong as its weakest point; a few hours at the wrong temperature can spoil a whole consignment. Perishables therefore demand refrigerated equipment, prompt handling, and temperature records as part of the shipment information.

Fragile cargo means goods susceptible to damage from impact, vibration, pressure, or improper handling. Glassware, ceramics, mirrors, electronic devices, precision instruments, and tiles are typical examples. Fragile goods rarely need special transport equipment; they need protection engineered into the packaging and care built into the handling. Inner cushioning absorbs shocks,

outer cartons resist compression, packages are marked so every handler knows to treat them gently, and stacks are kept to heights the bottom packages can bear.

Both categories teach the same lesson in different ways: the cargo's characteristics, not the carrier's convenience, set the conditions of carriage. A forwarder who accepts frozen tuna must be able to guarantee refrigeration from pickup to delivery; a warehouse storing ceramics must control how high the pallets are stacked. Storage conditions, including temperature, humidity, ventilation, and stacking limits, belong in the planning conversation too. Identifying special-handling cargo early, while the shipment is still being planned, prevents damage no paperwork can repair afterwards.

**KEY CONCEPT**

### Dangerous goods and the nine hazard classes

Dangerous goods are substances or articles that present hazards during transportation and are subject to applicable safety regulations. The category is broader than beginners expect, covering ordinary products: aerosol cans, lighter fuel, paints and thinners, bleach, fireworks, pool chemicals, and the lithium batteries inside phones and power banks. What makes an item a dangerous good is not that it is unusual, but that it presents a hazard in transport and is therefore regulated.

Internationally, dangerous goods are divided into nine classes, worth learning in outline.

1. Explosives.

2. Gases, including flammable, non-flammable non-toxic, and toxic gases. Aerosols travel in this class.

3. Flammable liquids, such as petrol, paints, and many solvents.

4. Flammable solids, which also covers materials liable to spontaneous combustion and materials that emit flammable gas in contact with water.

5. Oxidising substances and organic peroxides, which feed fires rather than burn alone.

6. Toxic and infectious substances.

7. Radioactive material.

8. Corrosive substances, which destroy skin, metal, and other materials.

9. Miscellaneous dangerous substances and articles, a class that includes lithium batteries, dry ice, and environmentally hazardous substances.

Each regulated substance or article is identified by a UN number, a four-digit code used worldwide, together with its proper shipping name; aerosols, for example, travel under UN 1950, and lithium-ion batteries under UN 3480. For packing purposes, most classes are

additionally graded into packing groups by degree of danger: packing group I for great danger, II for medium danger, and III for minor danger. Classes 1, 2, and 7 have no packing groups.

Two cautions frame everything else. First, dangerous goods are accepted for transport only under strict conditions of declaration, packaging, marking, and segregation, which the important note near the end of this lesson explains. Second, the full treatment, including the modal rulebooks and their documentation, belongs to Module 11 of this programme; here the aim is to recognise the classes and respect them.

**KEY CONCEPT**

### Project and oversized cargo

Not everything fits in a box. Project cargo is the collective name for items that are individually so large, heavy, or awkward that they cannot travel as ordinary units or in standard containers: electrical transformers, industrial generators, mining and road-building plant, prefabricated structures, boats, and factory machinery. A piece that exceeds standard container dimensions is called out-of-gauge cargo, and each such movement is planned almost as an individual engineering operation.

Standard container types give some help. Flat rack containers, with no sides or roof, accept machinery and vehicles that would never pass through a dry box's doors; open top containers, with a removable tarpaulin roof, take tall or long cargo that must be lowered in from above by crane. Beyond container limits, cargo travels on specialised trailers, cradles, and platforms, and ports lift it with heavy cranes.

What distinguishes project cargo from ordinary freight is the depth of planning. A single oversized movement may require checking the lifting capacity at each transfer point, confirming that the route can carry the weight and dimensions, obtaining any special permissions, and agreeing schedules with everyone involved, because a transformer that cannot pass under a bridge or be lifted by the available crane stops the project. The essential points are these: oversized cargo is identified early, planned individually, moved with specialised equipment, and marked so that everyone along the route knows its weight, balance, and lifting points. Ghana sees such movements regularly, when heavy construction, mining, and energy equipment arrives through the ports for projects inland.

**KEY CONCEPT**

### Packaging: levels, functions, and suiting the journey

Packaging means the materials and methods used to contain, protect, preserve, and prepare goods for handling, storage, and transportation. Packaging works in levels, easiest to see on an everyday product.

1. Unit or primary packaging. The bottle, jar, wrapper, or sack that holds the product itself: the jar of shea butter the customer buys. It is designed for the shopper, not mainly for the warehouse.

2. Inner packaging. The material inside the outer box that cushions and separates: cardboard dividers between jars, polythene liners that keep moisture out, foam inserts for electronics. Inner packaging protects items from one another and from the shocks of the journey.

3. Outer or transport packaging. The carton, sack, drum, crate, or pallet that handlers actually lift and stack. Transport packaging is designed for the logistics chain, and it carries the marks and labels the chain reads.

Whatever the level, packaging performs four functions.

1. Protect. Against mechanical shocks and vibration, crushing and stacking loads, moisture and humidity, contamination, and, where relevant, temperature extremes.

2. Unitise. Packaging gathers many small items into one handleable unit: twelve jars into one carton, twenty cartons onto one pallet, a thousand cartons into one container. Unitisation is what makes modern handling speeds possible.

3. Identify. The package surface carries the marks, numbers, and labels that tell the chain what the package is, where it goes, and how to treat it.

4. Comply. Some packaging is required by regulation rather than choice: dangerous goods need packaging certified to performance standards, and food products need hygienic packing.

Finally, packaging must suit the journey: a carton adequate for a short, dry air trip may fail on an ocean voyage of weeks through humid, salt-laden air, stacked beneath other cargo and lifted repeatedly between truck, warehouse, and ship. Matching packaging to the mode and to every handling step along the chain is a core logistics decision, not an afterthought.

**KEY CONCEPT**

### Common packaging forms and palletisation

Across warehouses and ports the same packaging forms appear again and again, each suited to particular goods.

1. Cartons. Corrugated boxes, the standard transport packaging for consumer goods, processed foods, and dried products. Cartons are light, printable, and stackable, and they are almost always unitised on pallets for movement in quantity.

2. Sacks. Bags of woven plastic, paper, or jute for free-flowing goods: cement, rice, sugar, fertiliser, and cocoa beans. Sacks are cheap and space-efficient but vulnerable to snagging, moisture, and overstacking, which is why their stacking height and covers matter so much.

3. Crates. Wooden or plastic framed boxes for heavy, valuable, or fragile items: machinery, tiles, glassware, and equipment. A crate can be engineered for the exact piece it carries and marked with lifting points and orientation arrows where needed.

4. Drums. Cylindrical containers of steel or plastic, commonly holding liquids: shea butter, oils, chemicals, and food ingredients. Drums seal tightly, stack reliably, and can be palletised for mechanical handling.

5. Big bags. Large woven sacks, often with lifting loops, each holding several hundred kilograms up to roughly a tonne of free-flowing goods such as granules, nuts, or powders. Big bags combine bulk economy with unit handling: a forklift lifts one pair of loops and moves what would otherwise be dozens of small sacks.

Unitising cartons, sacks, and drums leads to palletisation, the arrangement and securing of goods on pallets to facilitate handling, storage, and transportation. A pallet is a flat portable platform, usually wooden or plastic, that a forklift can slide beneath. Goods are stacked on it in a planned pattern, then secured with stretch wrap or strapping so the whole load behaves as one stable unit. The benefits are considerable: a forklift moves hundreds of cartons in seconds, packages are handled less often and therefore damaged less, stacks are more stable, and counting is quicker. The cost is the pallet itself and the space it occupies, a trade-off shippers accept worldwide.

**KEY CONCEPT**

## Cargo handling: principles and equipment

Cargo handling means the activities involved in loading, unloading, lifting, moving, stacking, and securing goods. Every package is handled many times between sender and receiver, and each handling is a chance to do the job well or badly. Good handling follows a few principles that apply from a village loading bay to a modern container terminal.

1. Respect the package's limits. Stacking limits, whether printed on the packaging or agreed for the product, exist because bottom layers fail when loads are excessive.

2. Observe orientation. Packages marked this side up must stay this side up; an inverted drum of liquid or carton of instruments can leak or break without a single knock.

3. Handle fragile cargo as fragile. No dropping, throwing, or dragging; cushioning stays in place; and fragile units are never used as a base for other cargo.

4. Keep loads balanced. When lifting, loading a truck, or stuffing a container, the centre of gravity, the point where the load's weight is considered to act, should stay low and near the middle. Unbalanced loads topple, stress equipment, and make transport dangerous.

5. Segregate incompatible goods. Some cargoes must never travel or sit together: dangerous goods away from foodstuffs, oxidising materials away from flammables, wet goods away from dry. Segregation is a planned decision made before loading, not an accident of the loading order.

6. Protect from the elements. Covers, dry floors, and sheltered storage keep rain, humidity, and sun from doing quiet damage between the handling steps.

Handling equipment matches the packaging: forklifts and pallet trucks need pallets or firm bases, port cranes lift containers and heavy pieces, conveyors move loose parcels, and container handlers stack boxes in yards. Where none of these reach, manual handling does, which is why safe lifting technique and clear floor discipline remain part of the basics. The planner's task is to match equipment to cargo: a drum needs a clamp, a big bag needs loop hooks, and a fragile crate needs steady hands.

**KEY CONCEPT**

## Cargo marking: the information on the package

Cargo marking means the application of identification details, handling instructions, or other relevant information to cargo packages. Marking turns an anonymous box into an addressed, instructable unit that anyone in the chain can handle correctly without opening it or reading the sales contract. Marks fall into three layers.

1. Identification marks. Who the package belongs to and where it goes: the consignee's mark or code, the destination port or city, the country of origin, the package number within the consignment, such as carton 137 of 1,000, and often the gross weight. On palletised and containerised cargo these details also appear on the shipping documents, a connection developed below and in the next lesson.

2. Handling instructions. Standard pictorial symbols and words that tell every handler how to treat the package: this way up, fragile, keep dry, do not stack, centre of gravity, stacking limit. Pictograms matter because a package may be handled by people who share no common language along its route; the picture of a wine glass or an umbrella needs no translation.

3. Regulatory labels. Labels required by regulation, above all the hazard labels and UN number markings for dangerous goods, which tell carriers, terminals, and emergency services exactly what hazard a package presents.

Good marking practice passes three tests. It is accurate, agreeing with the documents and the actual contents. It is durable, able to survive rain, abrasion, and handling without becoming unreadable. And it is honest: marks that mislead, whether by accident or design, put handlers and cargo at risk and create disputes months later.

Marking is best understood as the information layer of packaging: packaging protects the goods physically, marking protects them operationally, because a package that cannot be identified, oriented, or risk-assessed from its marks will sooner or later be handled wrongly.

**WORKED EXAMPLE**

### Packing a containerised export of shea butter products

A cosmetics producer in Accra has concluded a sale of shea butter products to a buyer in Rotterdam and must pack 1,000 cartons for ocean shipment through Tema. All figures in this example are illustrative for teaching purposes.

Step 1: The product and its packaging levels. Each carton holds 12 jars. The jar is the primary packaging, designed for the customer; a cardboard divider set inside each carton is the inner packaging, keeping the jars from touching; the carton itself is the transport packaging. A packed carton measures roughly 0.4 by 0.3 by 0.25 metres, about 0.03 cubic metres, and weighs about 12 kilograms.

Step 2: Palletisation. Staff stack 20 cartons on each wooden pallet in a block pattern and secure the load with stretch wrap. The 1,000 cartons fill 50 pallets, each now a single stable unit a forklift can move in seconds.

Step 3: Choosing the container. The planner compares the cargo against container capacity. Volume: 1,000 cartons at about 0.03 cubic metres come to about 30 cubic metres before the pallets beneath them are counted, which already fills the roughly 33 cubic metres inside a twenty-foot container, and fifty pallets cannot be fitted into one. Weight: 1,000 cartons at 12 kilograms give 12,000 kilograms, comfortably below a forty-foot container's payload of approximately 26,700 kilograms. The cargo is light for its volume, so a forty-foot box is right; the pallets travel in it double-stacked with room to spare, while a twenty-foot container would accept only part of the load.

Step 4: The marking scheme. Before wrapping and loading, staff mark every carton alike: the consignee's code and destination, RTM-256, ROTTERDAM; the origin, PRODUCT OF GHANA; the package number, from carton 1 of 1,000 through carton 1,000 of 1,000; and the handling pictograms, this way up and keep dry, because both contents and cardboard suffer from moisture and inversion.

Step 5: Loading and handover. The 50 pallets are stuffed into the container in a planned pattern, the load is secured, and the container is sealed. The container and seal numbers, the package count, and the marks are recorded for the shipping documents that will accompany the cargo.

What does this example demonstrate? Packaging worked in levels, from jar to divider to carton to pallet to container, each level adding protection and unitisation. The container choice followed from arithmetic against the container's known capacity. The marking scheme prepared the cargo to travel: any handler between Accra and Rotterdam can identify, orient, and deliver each carton without further explanation.

Key lesson: pack from the product outward, unitise for handling, match the container to the cargo's volume and weight, and mark the packages so that any handler, anywhere, can handle them correctly.

**WORKED EXAMPLE**

### A handling failure: bagged cocoa damaged on the road to Tema

Not every movement goes well, and failures teach more than successes. All figures in this example are illustrative for teaching purposes.

An exporter in Kumasi has sold 1,200 jute sacks of cocoa beans to a European buyer, for shipment through Tema Port. Each sack holds roughly 60 kilograms, so the consignment weighs about 72 tonnes and needs three truckloads for the journey south.

Step 1: The loading. To save time, the loading crew piles the sacks directly onto three open truck beds without pallets or dunnage, stacks them high to fit each load on one deck, and sets off without waterproof covers.

Step 2: What went wrong. Three failures compound one another. First, the stacks are too high: on the rough road the truck's flexing drives the top sacks' weight into the bottom rows, and by Tema the lowest sacks have burst, spilling beans that are swept up and rebagged in whatever material is at hand. Second, the sacks sit directly on the bare bed, so road spray and a midday rainstorm soak the lower layers; wet jute stains, and damp cocoa is at risk of mould, which the buyer will reject. Third, nothing separates the sacks from the bed, the straps, or one another, so abrasion opens seams along the stack.

Step 3: The consequences. At the port, the receiver's inspection notes burst and stained sacks, wet layers, and rebagged loose beans. The buyer raises a claim: part of the quantity is lost outright, more is discounted for suspected quality damage, and the exporter's recovery against the carrier is weakened because the way the goods were packed and loaded contributed to the loss. A loading job that saved an hour has cost a margin and a customer's confidence.

Step 4: What should have happened. The sacks should have travelled on pallets or at least on clean dunnage, stacked to a height the bottom sacks could bear, covered with waterproof sheeting for the whole journey, and repacked properly, not improvised on the dock, wherever damage appeared on the way.

What does this example demonstrate? Improper packaging and handling convert a sound product into a claim: overstacking crushed, moisture spoiled, and abrasion opened, each failure small alone but decisive together.

Lessons from the example: stacking limits, dry separation from the vehicle bed, and weather protection are not formalities; they are the difference between cargo that arrives saleable and cargo that arrives as a dispute.

**WORKED EXAMPLE**

### Dangerous goods under control: a consolidated shipment from Singapore

An international example shows how this lesson's rules come together. A freight forwarder in Singapore is consolidating a groupage consignment, meaning several customers' goods combined in one container, for ocean shipment to Rotterdam: 60 cartons of aerosol paints, 40 drums of a solvent-based cleaning fluid, and 200 ordinary cartons of plastic household goods. All figures in this example are illustrative for teaching purposes.

Step 1: Classification. The forwarder starts from each product's safety data sheet, the standard document in which the manufacturer describes the product's hazards. The aerosols, which hold flammable propellant under pressure, are dangerous goods of Class 2, gases; the cleaning fluid is a flammable liquid, Class 3; the plastic goods are not dangerous goods at all. Each dangerous product is matched to its UN number and proper shipping name, the aerosols under UN 1950, and the flammable liquid is assigned a packing group according to its degree of danger; gases, being a class without packing groups, are not graded in this way.

Step 2: Declaration. The forwarder declares the dangerous goods to the ocean carrier on the dangerous goods declaration, stating the UN numbers, proper shipping names, classes, packing groups, quantities, and packing. The carrier's acceptance staff check the declaration before the goods are accepted; undeclared dangerous goods discovered later can see the shipment refused or offloaded.

Step 3: Packaging, marking, and labelling. The dangerous goods travel in packaging certified to the required performance standards. Each package is marked with its UN number and proper shipping name and bears the hazard label for its class; the container itself displays the corresponding placards so that everyone who sees it knows a flammable cargo is inside.

Step 4: Segregation. In the consolidation plan the forwarder keeps the two classes of dangerous goods apart as the sea rulebook, the IMDG Code, requires, stows both away from the plastic household goods, and confirms nothing incompatible shares the space. Segregation is decided on paper before a single package is loaded.

Step 5: Carriage. The accepted, declared, labelled, and segregated container sails to Rotterdam, where receiving checks find marks and documents in agreement.

What does this example demonstrate? Dangerous goods do not travel by accident. Classification from the safety data sheet, declaration to the carrier, compliant packaging and labels, and planned segregation together make a hazardous consignment safe to carry; remove any one step and the hazard travels with the goods.

Key lesson: the declaration, packaging, and labels of dangerous goods transport are not bureaucracy laid over the cargo; they are the safety system itself.

**IMPORTANT NOTE**

## Why dangerous goods demand special acceptance

A dangerous package cannot simply be handed to a truck driver or stacked into a hold like any other carton. The reason is that dangerous goods threaten three things at once: the people who handle them, the vehicle, vessel, or aircraft that carries them, and the cargo stowed nearby. A leaking drum, an overheating battery, or a hidden aerosol crushed in a stack can start a fire, poison a crew, or destroy cargo worth many times the offending package's value. No part of the chain therefore treats dangerous goods as ordinary freight.

Acceptance is the gatekeeper moment. A carrier accepts dangerous goods only after they are properly declared, classified, packaged, marked, and, where required, segregated; for air transport, acceptance is performed only by trained and authorised staff working through detailed checklists. Sea and air each have their own rulebooks, and the declaration states the UN number, proper shipping name, class, packing group, and quantities, so that every later handler knows what is inside.

Two warnings complete the picture. First, dangerous goods hide inside ordinary-looking cargo: lithium batteries sit in countless electronic devices, and dry ice, used to keep samples frozen, is itself a Class 9 dangerous substance. Shippers must identify such items honestly rather than assume careful packing makes them ordinary. Second, misdeclaration is not a paperwork shortcut but a genuine danger: an undeclared hazard travels with no precautions, endangering people and cargo, and when discovered it brings refusal, penalties, and liability for any harm caused.

Module 11 treats dangerous goods in depth. At this stage, remember the principle: dangerous goods earn their special handling because the consequences of treating them casually fall on everyone in the chain.

**IMPORTANT NOTE**

## The same commodity can travel as different cargo

A common beginner error is to treat cargo type as a permanent property of the product, as if cocoa were simply bulk and machinery simply general cargo. In reality, classification follows the form in which the goods actually travel, and the same commodity can legitimately appear in several categories.

Consider the examples this lesson has used. Cement leaves the factory as bagged general cargo, while its raw material, clinker, crosses the oceans as dry bulk. Cocoa beans packed in jute sacks are general cargo, counted sack by sack; the same beans flowing loose through a bulk handling chain are dry bulk. Shea butter in sealed drums is packaged general cargo, drum by drum; the same butter pumped into a heated tank is liquid bulk. Even a carton of tinned tomatoes changes character at the container's doors: as individual cartons the consignment is general cargo; stuffed into a container, it is handled as containerised cargo.

The classification matters because everything operational hangs from it. The form of travel decides the packaging, the handling equipment, the storage arrangement, the documents, the rates, and the risks. Bagged cargo can be counted and split into small deliveries but costs handling at every move; bulk cargo moves cheaply in vast quantities but needs specialised terminals and resists subdivision.

So when planning or describing a shipment, ask not what the product is but how it will travel: in packages or loose, on pallets, in a container, in a hold, in a tank. Answer that, and the cargo type, with all its requirements, follows.

**IMPORTANT NOTE**

### Marks and labels connect cargo to its documents

Throughout this lesson, packages have carried marks: consignee codes, package numbers, pictograms, hazard labels. It is tempting to see these as decoration, but that understates their role. Marks and labels are the physical cargo's side of a conversation with the paperwork, and the conversation must agree.

When a consignment is booked and shipped, its identification marks, package counts, weights, container numbers, and seal numbers are recorded on transport documents. At every handover, the receiver checks the cargo against those documents by reading the marks: carton 137 of 1,000 is expected, counted, and signed for because the documents say it exists. Where marks and documents disagree, the result is delay while staff reconcile pieces, misdelivery when a package cannot be attributed to a consignment, or a disputed claim when the receiver cannot prove what was received. In customs processing too, the description and marks of packages must line up with the declaration that describes them.

Handling pictograms and hazard labels carry the same principle further: they put the handling instructions and the safety information where the cargo will be, not merely in an office file. A package whose fragility is recorded only in a document nobody at the loading bay has read will be handled as if it were unbreakable.

The practical discipline is consistency: mark the packages to match the documents, and the documents to match the packages, every time. The next lesson introduces logistics documents properly, and as each one appears, notice how the marks, numbers, and labels this lesson placed on the packages reappear on the paper. Cargo and documents tell one story, or the story falls apart.

## Practice questions (Lesson Checks)

**Practice Question 1**

A vessel berths at Tema Port carrying three consignments: maize poured loose into the ship's hold, cartons of tinned tomatoes stacked on pallets, and frozen fish packed inside refrigerated containers.

Which statement best describes how these cargoes are classified?

1. All three are general cargo because each consignment consists of food carried by sea.

2. All three are bulk cargo because each is carried in large quantities on the same vessel.

3. The loose maize is dry bulk cargo, the palletised cartons are general cargo, and the frozen fish is containerised cargo requiring temperature-controlled handling. ✓

4. The classification depends on the value of each product: the most valuable consignment is general cargo and the rest is bulk.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Classification follows the form in which each consignment travels, not the product's food use and not the fact that all three share a vessel. The maize is poured loose into the hold with no packaging, which is the definition of dry bulk cargo. The tinned tomatoes travel as individual cartons stacked on pallets, countable units with their own packaging and marks, which makes them general cargo. The frozen fish moves inside refrigerated containers under temperature control, so it is containerised cargo with a perishable, special-handling requirement.

Answer 1 fails because belonging to the same product family does not make cargoes alike; the packaging and form of travel decide the classification. Answer 2 fails because bulk describes unpackaged cargo in direct contact with the hold or tank, which applies only to the maize. Answer 4 fails because price plays no part in classification; a cheap commodity in sacks and a costly one in a hold are still classified by their form of travel.

Takeaway: to classify cargo, look at how it travels, not at what it is.

**Practice Question 2**

An exporter in Accra wants to ship 200 cartons of aerosol air fresheners by sea. A colleague suggests declaring them as ordinary cartons of household goods because they are packed carefully and stacked securely.

Which option best explains why this suggestion is wrong?

1. Declaring the aerosols as ordinary goods would be acceptable cost management, since the classification of a consignment can be chosen to reduce freight charges.

2. Dangerous goods may never be transported by sea, so the consignment of aerosols cannot be shipped at all.

3. Aerosols are dangerous goods of Class 2, gases, so they must be declared with their UN number and proper shipping name, packed and labelled to the applicable requirements, and accepted by the carrier; careful packing does not remove the hazard. ✓

4. Classification follows the buyer's preference, so the exporter may describe the goods however the customer requests.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Aerosols contain propellant under pressure and are classified as dangerous goods of Class 2, gases, so they fall under the dangerous goods requirements however carefully they are packed. Transporting them legally requires declaring them to the carrier with their UN number and proper shipping name, packing them in compliant packaging, applying the hazard labels, and obtaining the carrier's acceptance. Careful stacking is not among the requirements, because it does nothing to control the hazard itself.

Answer 1 fails because the classification of dangerous goods is set by regulation for safety reasons, not chosen for cost. Answer 2 fails because dangerous goods are carried by sea every day; the rules govern how they travel, they do not forbid them. Answer 4 fails because classification follows the hazard of the contents, not the parties' preference; misdeclaring the goods would endanger the vessel and crew and expose the exporter to refusal and penalties.

Takeaway: if a product presents a transport hazard, it must enter the dangerous goods system; packing quality never substitutes for declaration.

**Practice Question 3**

A company in Accra must send 300 cartons of glassware to a customer in Tamale. The journey involves loading at the warehouse, a long road journey, unloading, and storage in a warehouse where other goods will be stacked nearby.

Which option describes the most suitable basic packaging and handling approach?

1. Loose cartons loaded directly onto the truck bed to save space, since glassware does not need any special treatment once packed.

2. Cartons with inner cushioning, stacked on a wrapped pallet, marked this way up and fragile, lifted by forklift, and stacked only to a safe height in the warehouse. ✓

3. Each glass wrapped in paper and the cartons then piled as high as the warehouse allows, because stacking height does not affect packaged goods.

4. Plain cartons without marks or pictograms, because handling instructions are useful only for dangerous goods.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Glassware is fragile cargo, so the packaging must absorb the shocks of the road and the handling chain, and the handling must respect the marks. Inner cushioning protects each piece, the palletised and wrapped cartons move as stable units, the this-way-up and fragile marks instruct every handler along the route, and a safe stacking height in the warehouse keeps the bottom cartons within the compression they can bear.

Answer 1 fails because loose cartons on a truck bed take the full vibration of the journey and the full weight of any stacking, exactly the pressures fragile goods cannot resist. Answer 3 fails because packaging does not remove stacking limits; compression crushes the bottom layers of even well-packed cartons when stacks are too high. Answer 4 fails because handling marks exist for fragile, perishable, and orientation-sensitive goods generally, not only for dangerous goods; a glassware carton without its pictograms will be handled like any anonymous box.

Takeaway: suitable packaging, correct marks, and disciplined stacking form the basic formula for moving fragile cargo safely.

### Fact-check appendix (audit notes — not for entry)

The definitions of cargo, general cargo, breakbulk cargo, bulk cargo, containerised cargo, perishable cargo, fragile cargo, dangerous goods, packaging, cargo handling, palletisation, and cargo marking follow the lesson brief's key terms. — source: curriculum/lesson_briefs/module-03/lesson-3.3.md, key terms from curriculum.

Standard containers are built to ISO 668; a twenty-foot dry container has internal dimensions of approximately 5.898 x 2.352 x 2.393 metres (about 33 cubic metres), a maximum gross mass of 30,480 kilograms, tare of approximately 2,230 kilograms, and payload of approximately 28,250 kilograms. — source: research sheet trade-transport-standards.md, section 6 (Containers, ISO 668 / ISO 1496).

A forty-foot dry container has internal dimensions of approximately 12.032 x 2.352 x 2.393 metres (about 67.5 cubic metres) and payload of approximately 26,700 kilograms; a forty-foot container roughly doubles the volume but not the permitted weight of a twenty-foot. — source: research sheet trade-transport-standards.md, section 6.

A forty-foot high-cube container has an external height of 9 feet 6 inches (2.896 metres) and internal height of approximately 2.698 metres, quoted in the lesson as approximately 2.7 metres. — source: research sheet trade-transport-standards.md, section 6.

Tare is the empty container weight, payload the maximum permissible cargo weight, and maximum gross the safety-rated total of tare plus cargo. — source: research sheet trade-transport-standards.md, section 6.

TEU counts twenty-foot and FEU forty-foot containers; 1 FEU equals 2 TEU; a forty-foot high cube still counts as one FEU. — source: research sheet trade-transport-standards.md, section 6.

Special container types: reefer (refrigerated, insulated, for perishables), open top (removable tarpaulin roof, crane-loaded over-height or long cargo), flat rack (no sides or roof, for out-of-gauge machinery), and ISO tank (tank in a twenty-foot frame for liquids, gases, and powders, often dangerous goods). — source: research sheet trade-transport-standards.md, section 6.

The UN Model Regulations divide dangerous goods into nine classes: 1 Explosives; 2 Gases (flammable, non-flammable non-toxic, toxic); 3 Flammable liquids; 4 Flammable solids (including spontaneously combustible and water-reactive materials); 5 Oxidising substances and organic peroxides; 6 Toxic and infectious substances; 7 Radioactive material; 8 Corrosive substances; 9 Miscellaneous dangerous substances and articles, including lithium batteries, dry ice, and environmentally hazardous substances. — source: research sheet trade-transport-standards.md, section 9 (Dangerous goods).

UN numbers identify each regulated substance or article, four digits with the UN prefix, together with its proper shipping name; for example UN 1950 aerosols and UN 3480 lithium-ion batteries. — source: research sheet trade-transport-standards.md, section 9.

Packing groups grade danger for packing performance: PG I great danger, PG II medium danger, PG III minor danger; they apply to most classes except 1, 2, and 7. — source: research sheet trade-transport-standards.md, section 9.

The shipper declares dangerous goods on the modal declaration stating UN number, proper shipping name, class, packing group, quantities, and packing compliance; packaging must meet UN performance standards; hazard labels are applied to packages and placards to units or vehicles as the mode requires. — source: research sheet trade-transport-standards.md, section 9.

The IMDG Code is the sea dangerous goods rulebook; the IATA DGR governs air transport. — source: research sheet trade-transport-standards.md, section 9.

Air cargo dangerous goods are accepted only by trained, authorised staff working against the IATA DGR checklists. — source: research sheet trade-transport-standards.md, section 8 (Air cargo, acceptance basics).

A safety data sheet is the manufacturer's document classifying a product's hazards and is the starting point for deciding whether goods are regulated for transport. — source: research sheet trade-transport-standards.md, section 9 (SDS/MSDS).

Cold-chain logistics means temperature-controlled storage and transport for perishables and pharmaceuticals with continuous monitoring. — source: research sheet logistics-scm-frameworks.md, section 1.

Groupage means combining several shippers' goods into one consignment or container. — source: research sheet logistics-scm-frameworks.md, section 6 (consolidation, FTL/LTL/groupage).

Standard handling pictograms (fragile, keep dry, this way up, centre of gravity) and common handling equipment (forklifts, pallet trucks, cranes, conveyors, container handlers) are general industry practice. — source: general business/industry knowledge.

Big bags are large woven bags with lifting loops, typically holding several hundred kilograms up to roughly one tonne of free-flowing goods. — source: general industry knowledge.

Ghanaian illustrations (bagged cement distribution, jute-sacked cocoa, drummed shea butter, frozen tuna for export, heavy mining, construction and energy equipment arriving by sea) are illustrative teaching examples drawn from general knowledge of Ghanaian trade, with no institutional claims attached. — source: general knowledge, illustrative.

All quantities and arithmetic in the worked examples are illustrative for teaching purposes, as stated within each example. — source: teaching convention per WRITING_SPEC.md rule 5.1.

# Lesson 3.4 — Introduction to Logistics Documents

## Summary

This lesson introduces the basic documents used to identify, describe, transport, and deliver goods during logistics operations. Learners will explore the purposes of commercial invoices, packing lists, transport documents, delivery notes, and other supporting records, while understanding the importance of accurate documentation in shipment coordination and cargo movement.

## Learning objectives

- Explain the purpose of documentation in logistics operations.
- Identify common documents used during the movement of goods.
- Describe the basic functions of commercial invoices and packing lists.
- Recognise the purpose of bills of lading, air waybills, and road transport documents.
- Explain the role of delivery notes and proof-of-delivery records.
- Identify common shipment details recorded in logistics documents.
- Explain why accurate and consistent documentation is important.
- Recognize basic documentation errors that may cause shipment delays or discrepancies.

## Key terms

**Logistics Documentation —** The documents and records used to support, control, and verify logistics activities.

**Commercial Invoice —** A document issued by a seller that provides details of goods sold, their value, and the commercial transaction.

**Packing List —** A document describing the contents, quantities, packaging, weights, or dimensions of goods in a shipment.

**Bill of Lading (B/L) —** A maritime transport document issued by or on behalf of a carrier that acknowledges receipt of goods and provides evidence of the contract of carriage; certain forms also function as documents of title.

**Air Waybill (AWB) —** A non-negotiable air transport document that provides evidence of the contract of carriage and acknowledges receipt of cargo for air transportation.

**Consignment Note —** A transport document containing information about goods entrusted to a carrier for delivery.

**Delivery Note —** A document accompanying goods that identifies the items and quantities being delivered.

**Proof of Delivery (POD) —** A document or electronic confirmation showing that a shipment has been delivered or received.

**Shipment Reference Number —** A unique identifier assigned to a shipment or logistics transaction for tracking and recordkeeping.

**Document Discrepancy —** An inconsistency, omission, or error identified within a document or between related documents.

## Teaching content

**KEY CONCEPT**

### Why goods movement needs documents

Think about collecting a parcel from a delivery office. You do not simply reach behind the counter and take the box. You identify yourself, and you sign a record showing that you received the parcel. That small everyday exchange contains the whole idea of this lesson: goods cannot move far without records moving with them.

Logistics works the same way, on a much larger scale. A shipping line will not load a container on trust. A warehouse will not release goods because a driver says they are expected. Customs will not release an import because a trader promises that the goods are correctly described. In every case, a document does the talking.

Logistics documentation is the documents and records used to support, control, and verify logistics activities. Behind it lies a simple principle: goods move physically, but information must move ahead of the goods and with them. Before a refrigerated container of frozen fish leaves a cold store in Tema, information about it must already have reached the shipping line, the port and the buyer, and while the goods travel, their documents travel too.

The main purposes of logistics documentation are the following.

1. Instructions and contracts
Documents tell every participant what to do and record what was agreed. A dispatch instruction tells a warehouse what to load and where to send it, and a transport document records the agreement under which a carrier moves the goods.

2. Receipts
Documents confirm that goods were actually handed over or received, protecting both sides with a record of the handover.

3. Evidence of the right to claim the goods
On some shipments, holding the right document is what allows a person to claim the goods at destination. The document controls who may collect them.

4. Declarations to authorities
Documents declare to customs and other authorities what the goods are, what they are worth and where they come from, so that the correct rules can be applied.

5. Records supporting payment
Invoices and delivery confirmations allow the seller to bill the buyer, and allow the buyer to verify that what it is asked to pay for was actually delivered.

One shipment may involve many documents, but they belong to a small number of families, and the next block introduces them.

**KEY CONCEPT**

## The families of logistics documents

Although a single international shipment can generate a thick file of paperwork, nearly every document in the file belongs to one of five families, and each family answers one question about the shipment.

1. Commercial documents describe the sale that the goods carry out. The main examples are the commercial invoice, the pro forma invoice and the packing list.

2. Transport documents describe the carriage of the goods. The bill of lading covers sea transport, the air waybill covers air transport, and the road consignment note covers road delivery. Each is issued when the carrier receives the goods.

3. Certificates provide independent confirmation of particular facts about the goods, such as where they come from or their state of health. Common examples are the certificate of origin, phytosanitary and health certificates, and quality certificates.

4. Customs declarations declare the goods to the customs authorities of the countries involved, so that the correct customs rules and charges can be applied.

5. Delivery records close the journey. The delivery note travels with the goods to the receiver, and the proof of delivery confirms that the handover happened.

The families work together. The customs declaration is prepared from the details on the commercial invoice and the packing list, the transport document links the goods to the carrier, and the delivery records prove that the goods arrived. Notice also that the same core details, such as quantities, weights and descriptions, appear again and again across the families. That repetition is deliberate, because different participants read different documents, and it is also why errors spread so easily, a point this lesson returns to.

One placement note before we continue. This lesson teaches what each document is for and when it appears. The deep treatment of every document, including how they are examined, corrected and matched, belongs to Module 6, which is devoted entirely to logistics documentation. Everything here is the foundation that module will build on.

KEY CONCEPT

## The details logistics documents record

Although logistics documents serve different purposes, most record a similar core of information about the shipment. Learning this core now makes every later document easier to read, because you will know what to look for.

1. The parties
The names and addresses of the sender and the receiver. In international trade the sender is known as the shipper and the receiver as the consignee. Documents may also name the agents or carriers involved in the movement.

2. Description of the goods
What the goods are, described clearly enough to identify them: frozen shrimp, ceramic floor tiles, diesel engine spare parts. Vague descriptions such as general goods invite questions, because nobody can verify what is really being moved.

3. Quantities and packages
How many cartons, bags, drums or pallets, and how many units each package contains. Quantities are the most commonly checked details in logistics, so they must be identical everywhere they appear.

4. Weights and measurements
The gross weight including packaging, the net weight of the goods themselves, and where relevant the volume or dimensions of the packages. Carriers need weights to plan loads safely, and authorities use them to verify declarations.

5. Origin and destination
Where the goods start their journey, where they are going, and the specific points where they are loaded and discharged.

6. Dates
When a document was issued and when the goods were dispatched, picked up or delivered. Dates create the timeline every participant works to.

7. Marks, numbers and references
Packages carry markings that identify them, and each shipment normally carries a shipment reference number, a unique identifier assigned to a shipment or logistics transaction for tracking and recordkeeping. The reference is how everyone involved points to the same shipment without confusion, in conversations, emails and computer systems.

8. Values and charges
The value of the goods and, where applicable, the freight charges, including whether the transport cost has been paid or remains payable at destination.

9. Handling instructions
Directions such as keep refrigerated or fragile, which tell everyone who touches the cargo how to treat it.

Because the same details appear on many documents, an error in one is easily copied into the others. Logistics professionals therefore check documents against each other, not against memory.

**KEY CONCEPT**

## Commercial invoices

The commercial documents describe the sale that the movement of goods carries out. The most important is the commercial invoice, a document issued by a seller that provides details of goods sold, their value, and the commercial transaction.

A commercial invoice typically shows the seller and the buyer, a description of the goods, the quantities, the unit prices and the total value, the currency, and the references linking the invoice to the order and the shipment. It is much more than a request for money. It performs three jobs at once.

1. It records the commercial transaction
The invoice is the seller's formal statement of what was sold, to whom, at what price and under what conditions. If a disagreement arises about what was agreed, the invoice is central evidence.

2. It supports payment
Buyers use the invoice to arrange payment, and where banks are involved in international payment arrangements, the banks examine the invoice together with the other documents. Payment flows through documents, and the invoice leads them.

3. It supports customs valuation
Customs authorities decide duties and taxes on the basis of the declared value of the goods, and the commercial invoice is the document that declares that value. An invoice that states a value which is not true misleads the authority and puts the shipment at risk.

A related document is the pro forma invoice, an advance copy of an invoice issued before the goods are dispatched. The seller uses it to set out the proposed details of the sale, and the buyer uses it to confirm the terms, arrange payment or apply for permits. The pro forma invoice is a quotation in invoice form, not a demand for payment. The commercial invoice, by contrast, is issued for the actual shipment and demands payment for goods actually sold.

KEY CONCEPT

## The packing list

The packing list, a document describing the contents, quantities, packaging, weights, or dimensions of goods in a shipment, is the document that describes the physical consignment. Where the commercial invoice describes the sale, the packing list describes the boxes.

A packing list for a consignment of 500 cartons shows how the goods are packed: how many cartons there are, how many units each carton contains, the gross and net weights, and where relevant the dimensions of each carton and the way the cartons are arranged on pallets. It carries no prices. Its purpose is physical, not financial.

Three groups of people rely on it.

1. Warehouse staff
They pick, count and pack against the packing list, and the receiving warehouse at the other end counts incoming cartons against it.

2. Carriers and terminal staff
They use the package counts, weights and dimensions to plan loading, stowage and handling, especially where goods must be kept refrigerated or stacked within limits.

3. Customs authorities
When they examine a consignment, they compare what is physically there against the packing list and the declaration to confirm that the goods match the paperwork.

Because the packing list and the commercial invoice describe the same goods from two angles, the two must agree completely. If the invoice says 500 cartons and the packing list says 550, one of them is wrong, and until the difference is resolved the shipment carries a question mark. How such disagreements arise, and what they cost, is treated later in this lesson.

KEY CONCEPT

## The bill of lading: the key document of sea transport

Sea transport has its own key document. A bill of lading is a maritime transport document issued by or on behalf of a carrier that acknowledges receipt of goods and provides evidence of the contract of carriage; certain forms also function as documents of title. The abbreviation B/L is standard in daily use.

The bill of lading is the classic example of one document serving all three of the roles a transport document can play.

1. Receipt for the goods
When the carrier receives the container or cargo and, in the usual case, loads it on board the vessel, the bill of lading records what was received and its apparent condition. It is the shipper's proof that the goods were handed over.

2. Evidence of the contract of carriage
The document sets out the agreement between the shipper and the carrier: the parties, the vessel and the ports, and the basis on which the goods are carried. If a dispute arises about the carriage, the bill of lading is the evidence of what was agreed.

3. Document of title
This is the function that makes the bill of lading famous. For certain forms of the document, possession of the properly issued original controls who may claim delivery of the goods at destination. That is why the document can be used to support international payment arrangements, and why goods travelling under it can even be sold while still at sea.

No other common transport document combines all three functions. One further point completes the sea picture: sea transport also uses a simpler, non-negotiable document called a sea waybill, which is a receipt and evidence of the contract but not a document of title. When and why traders choose one or the other is a Module 6 topic. For now, hold on to the three functions.

**KEY CONCEPT**

## Air waybills and road consignment notes

Air and road transport use documents that are simpler in one important respect: they are never documents of title.

The air waybill, abbreviated AWB, is a non-negotiable air transport document that provides evidence of the contract of carriage and acknowledges receipt of cargo for air transportation. It is the air cargo counterpart of the bill of lading in its first two functions: it proves the airline received the cargo, and it evidences the agreement under which the cargo is carried. It also serves as the freight bill and supports customs processing. What it never does is control who may claim the goods. The cargo is delivered to the consignee named on the document, and no original document has to be presented to collect it. Where a freight forwarder consolidates several shippers' goods into one air consignment, the airline issues a master air waybill for the consolidated load and the forwarder issues a house air waybill to each shipper for its own part of it; that structure is a Module 6 detail.

Road transport uses the consignment note, a transport document containing information about goods entrusted to a carrier for delivery. Like the air waybill, it is a receipt for the goods and evidence of the contract of carriage, and it is non-negotiable: it is not a document of title. The note normally accompanies the goods on the journey, so that the driver can show what the vehicle is carrying. In Europe, international road movements use a standardised form of consignment note under the CMR convention. Ghana is not part of that convention, and road movements in Ghana and the wider region travel under consignment notes issued under national and regional practice. The function, however, is the same everywhere: a receipt, a record of the carriage agreement, and a description of the goods travelling with the vehicle.

KEY CONCEPT

## Certificates that travel with goods

Some facts about goods cannot simply be declared by the trader; someone independent must confirm them. Certificates are documents in which an authority or an independent inspector states that a particular fact about the goods is true. Which certificates a shipment needs depends on the goods and on the countries involved. This lesson covers the names and purposes only.

1. Certificate of origin

This certificate states where the goods come from, that is, the country in which they were produced or manufactured. It matters because customs treatment often depends on origin. A non-preferential certificate supports ordinary customs treatment. A preferential certificate evidences that the goods qualify under a specific trade agreement, such as the ECOWAS trade liberalisation scheme or the African Continental Free Trade Area, and supports a claim to reduced or zero duty. The distinction is commercial: a preferential certificate can be worth money to the importer.

2. Phytosanitary certificate

For plants and plant products, the plant-protection authority of the exporting country may certify the phytosanitary condition of the goods, meaning their condition with respect to plant pests and diseases.

3. Health or sanitary certificate

For products of animal origin, such as meat or dairy, a corresponding health certificate may be required to confirm that the goods meet health requirements.

4. Quality or inspection certificates

An independent inspector may certify the quality, quantity or condition of the goods, for example at loading. These are common where buyers purchase large commodity parcels they cannot inspect personally.

Notice the common thread. Each certificate is issued by a party other than the buyer or the seller, which is precisely what makes it credible. A trader asserting that its goods are disease-free proves nothing; the exporting country's authority certifying it does.

KEY CONCEPT

## Customs declarations

The fourth family is the document that speaks to government: the customs declaration. When goods cross a border, the customs authority of the country must be told what the goods are, what they are worth, where they come from and where they are going, so that the authority can apply the correct rules, controls and charges. The declaration is the trader's formal statement of those facts, and it is prepared from the other documents in the file: the description and quantities come from the invoice and the packing list, the departure and arrival details come

from the transport document, and any certificates are presented alongside it. It is the clearest example of the document families working together.

In Ghana, the customs declaration for an imported consignment is lodged electronically in ICUMS, the Integrated Customs Management System, which operates as Ghana's single-window platform for trade declarations. In practice the lodging is done by a licensed clearing agent acting for the importer, and every consignment receives a Unique Consignment Reference, abbreviated UCR, which is used to track it through the process. The UCR is worth noticing. It is a real, national-scale example of the shipment reference number introduced earlier in this lesson, doing exactly what shipment references always do: giving every participant one name for one shipment.

Customs declarations are only introduced here. What a declaration contains, how customs examines and clears it, and what happens when something goes wrong are treated in the later modules of this course. The point to carry forward is simpler. The declaration is where all the other documents meet the authorities, and its accuracy depends entirely on theirs.

**KEY CONCEPT**

## Delivery notes and proof of delivery

The journey is not complete when the goods arrive; it is complete when the arrival is recorded. Two documents do this recording.

The delivery note is a document accompanying goods that identifies the items and quantities being delivered. It travels with the consignment, and it is the first document the receiver reads. The receiving process follows the same pattern everywhere. The receiver counts the packages, checks them against the delivery note, inspects their condition, and then signs. The signature says: what this document describes is what I received. Anything unusual, whether a shortage, damage or a wrong item, is noted on the document at the moment of delivery, while both parties are still present.

The proof of delivery, abbreviated POD, is a document or electronic confirmation showing that a shipment has been delivered or received. In its simplest form it is the signed delivery note itself. In modern operations it may be an electronic record, such as a captured signature, a photograph of the delivered goods or a confirmation in a tracking system. Its role is the same in every form: it is the evidence that the journey ended correctly, and it is what closes the shipment's file.

Proof of delivery also completes the payment loop. Before a buyer pays an invoice, the buyer's people match three records: the order, the delivery record and the invoice. This check is known as the three-way match, and it works only if the delivery records are accurate, which is why the moment of signing matters so much. A signed delivery note and proof of delivery for goods that were never counted is not a record. It is a guess with a signature on it.

**WORKED EXAMPLE**

## Following one export: frozen shrimp from Tema to Amsterdam

A seafood processor in Tema sells 500 cartons of frozen shrimp to a distributor in Amsterdam, travelling by sea in one refrigerated container. All figures are illustrative for teaching purposes. Follow the documents, stage by stage.

Step 1: The sale is agreed

The buyer receives a pro forma invoice setting out the proposed sale details and uses it to confirm the terms. The processor then issues the commercial invoice for the actual shipment: the parties, the goods, the quantity, the value and the currency.

Step 2: The goods are prepared

The shrimp are packed into 500 cartons of 10 kilograms net each and loaded into the refrigerated container. The packing list records the carton count, units per carton, weights and dimensions, and warehouse staff pick and count against it.

Step 3: The certificates are obtained

Because shrimp are products of animal origin, a health certificate confirming they meet health requirements is obtained from the competent authority, and a certificate of origin is prepared for the buyer's import formalities.

Step 4: Declaration and handover to the carrier

The processor declares the goods to customs in the required export formalities before loading. The container is delivered to the shipping line at Tema, and the carrier issues the bill of lading on receiving it: the processor's receipt, evidence of the contract of carriage, and the document of title controlling who may claim the container in Amsterdam.

Step 5: The information travels ahead

Copies of the invoice, packing list, certificates and transport document reach the buyer and its agent long before the vessel does. The documents have moved ahead of the goods.

Step 6: Arrival and clearance in Amsterdam

The buyer's agent claims the container with the bill of lading and lodges the import declaration with the Dutch customs authority, presenting the invoice, packing list and certificates. Customs checks that the documents tell one consistent story before release.

Step 7: Final delivery and record

When the distributor receives the shrimp, a delivery note accompanies them and the distributor signs for the 500 cartons, creating the proof of delivery that closes the file.

What does this example demonstrate?

Every stage produced a document, and each proved something specific: the invoice the sale, the packing list the physical contents, the certificates the origin and health status, the bill of lading

the handover and the carriage contract, and the delivery records the final receipt. Together they tell one consistent story.

Key lesson
Goods cannot complete an international journey on their own. They travel inside a set of documents assembled in order and kept consistent, each existing because someone, whether carrier, buyer, bank or authority, must be able to trust a fact without seeing the goods.

**WORKED EXAMPLE**

### An air shipment: spare parts from Singapore to Rotterdam

Documents adapt to the mode of transport. A machine manufacturer near Rotterdam has a production line stopped, waiting for a replacement control unit. The unit is in stock at the supplier's warehouse in Singapore. It is small, valuable and urgently needed, so it will travel by air. All figures in this example are illustrative for teaching purposes.

Step 1: Order and invoice
The manufacturer confirms the order, and the Singapore supplier issues a commercial invoice describing the control unit, its quantity and its value, with the references linking it to the order.

Step 2: Packing
The unit is packed into one wooden crate. The packing list records one crate, the gross weight of 85 kilograms and the dimensions. These details matter particularly in air transport, where weight and space determine the freight.

Step 3: Booking and the air waybill
The supplier's freight forwarder, the intermediary that arranges shipments on behalf of shippers, books space with an airline. When the airline receives the crate, an air waybill is issued naming the manufacturer as consignee. The air waybill acknowledges receipt of the cargo and evidences the contract of carriage, and it is non-negotiable: it is not a document of title.

Step 4: Release on arrival
At the destination airport there is no original document to present and no title to transfer. The airline's agent contacts the named consignee, the manufacturer's clearing agent completes the import formalities, and the cargo is released. Compare this with the ocean shipment in the previous example, where the bill of lading controlled who could claim the container. Air cargo moves to the named consignee because the air waybill says so, and nothing more is needed.

Step 5: Delivery and record
The crate is delivered to the manufacturer's factory, the delivery note is checked and signed, and the proof of delivery completes the file.

What does this example demonstrate?
The document families are the same by every mode, meaning an invoice, a packing list, a

transport document and delivery records, but the transport document changes character with the mode. Sea transport can use a document of title; air transport cannot.

Key lesson
When the mode changes, ask what the transport document can and cannot do. The air waybill is fast and simple precisely because it is not a document of title, and a seller consigning air cargo has no paper to control who collects it, which is why such shipments suit trading relationships built on trust.

**WORKED EXAMPLE**

### A domestic delivery and a five-carton shortage

Documentation discipline matters just as much on a domestic delivery inside Ghana. Consider a distributor that supplies cooking oil from its Tema warehouse to supermarkets in Accra. All figures in this example are illustrative for teaching purposes.

Step 1: Dispatch
A supermarket branch orders 240 cartons of cooking oil. The distributor's warehouse picks and loads 240 cartons onto a truck and prepares a delivery note identifying the items and the quantity of 240 cartons. The documents travel with the driver.

Step 2: Arrival and check
At the branch, the receiving staff do not sign on sight. They count the cartons as they are unloaded, and they count 235. They inspect the condition of the cartons as well. The delivery note says 240; the truck is holding 235.

Step 3: Recording the discrepancy
The staff record the actual count of 235 and note the shortage of 5 cartons on the delivery documents before signing. The driver countersigns, so both parties leave with the same honest record: 235 received, 5 short. The proof of delivery therefore shows what was really received, not what was planned.

Step 4: Investigation
The distributor now investigates with evidence in hand. Was a carton left at the warehouse? Did the loading count slip? Was a carton damaged and removed along the way? The warehouse dispatch records, the truck's load and the delivery note can be compared precisely because the delivery note records the truth.

Step 5: The payment follows the record
The commercial invoice will eventually claim payment for 240 cartons, but the supermarket's records will not agree with it: the order says 240, the delivery record says 235 and the invoice says 240. This three-way match is exactly the check that prevents the supermarket from paying for cartons it never received until the difference is resolved.

What does this example demonstrate?
The documents did their job because they were allowed to record reality. A shortage discovered and recorded at delivery is a small, claimable and investigable event. The same shortage discovered weeks later, or covered by an inaccurate signature, becomes an argument with no evidence.

Key lesson
Sign for what you counted, not for what the paper says. The delivery note and the proof of delivery protect both sides only when they record the truth at the moment of handover.

**IMPORTANT NOTE**

### One shipment, one story: documents must agree

The single most important quality principle in logistics documentation is consistency. Every document in a shipment's file is a retelling of the same story, and the details must agree from document to document: the quantities on the invoice, the packing list and the declaration; the weights on the packing list and the transport document; the descriptions everywhere; the parties and their names and addresses; and the references linking order to invoice to shipment.

A document discrepancy is an inconsistency, omission, or error identified within a document or between related documents. Discrepancies are common, and they are expensive. A quantity that disagrees stops a customs officer from verifying a declaration. A misspelled consignee name can leave cargo standing at a destination because nobody can match the paperwork. A missing date or signature can invalidate a document that a bank or an authority is being asked to rely on. In international payment arrangements, where banks pay against documents rather than goods, a single discrepancy can pause payment until it is corrected.

The basic errors to watch for are these: quantities that differ between documents; weights that do not match the packages; vague or incomplete descriptions of the goods; the wrong consignee or delivery address; missing dates, signatures or stamps; and references that do not link the documents to each other.

None of these errors is difficult to prevent, which is exactly the point. They are prevention problems, not accidents, and careful checking of one document against another before dispatch catches most of them.

Module 6 treats this subject properly: how documents are examined, how discrepancies are managed and corrected, and which consistency checks professionals run. For now, adopt the habit this lesson recommends: check documents against each other, never against memory.

**IMPORTANT NOTE**

### Never issue or sign a document with unverified details

Every document in this lesson makes claims about the real world: that 500 cartons exist, that goods came from a particular country, that 240 cartons were delivered. The entire system runs

on those claims being true, which is why the professional rule is absolute. Issue a document only when its details have been verified, and sign only for what you have personally counted or checked.

Your signature on a document is a statement of fact. When you sign a delivery note for 240 cartons without counting, you are certifying a receipt you did not verify. When a document is issued describing goods, values or origins that nobody checked, every downstream participant, whether carrier, bank or customs authority, builds its decisions on an unverified claim.

Integrity goes further than accuracy. Deliberately misstating a value, an origin, a quantity or a description on a document is misdeclaration. It can be done to reduce charges, to evade a restriction or to disguise a shortage, and wherever it is done it puts the trader at serious risk: goods can be held or seized, penalties can follow, and the trust that took years to build disappears. A document with false details is worse than no document at all, because it becomes evidence against the very people who created it.

The practical habits are simple. Check documents against the physical goods, not against assumption. Correct errors openly, on the record, so both parties see the correction. Report discrepancies promptly instead of hoping they will be forgotten. Accuracy and integrity are what make documents trustworthy, and trustworthy documents are what trade runs on.

## Practice questions (Lesson Checks)

**Practice Question 1**

A shipping line at Tema will not accept a 20-foot container of goods for loading until the exporter's booking is confirmed and the required documents are in order, even though the container is already standing at the terminal.

Which option best explains why logistics operations depend on documents in this way?

1. Because documents exist mainly to satisfy banks, and a shipment without them can only proceed when payment has already been made in advance.

2. Because documents carry the information ahead of and with the goods, serving as instructions, receipts, evidence for claiming goods, declarations to authorities and records supporting payment, so no participant can act correctly without them. ✓

3. Because documents are needed only when goods cross a border, and domestic movements can legally proceed without any records at all.

4. Because documents allow the carrier to inspect the physical contents of every sealed container before it is loaded onto the vessel.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Documents carry the information that logistics runs on. The lesson opened with the principle that goods move physically, but information must move ahead of and with them, and it identified five purposes that documents serve: instructions and contracts, receipts, evidence of the right to claim goods, declarations to authorities, and records supporting payment. A container standing at a terminal without its documents cannot be loaded, because the carrier has no verified instruction, no record of what it is receiving, and nothing to show the authorities.

Answer 1 narrows documents to banking, but payment is only one of the five purposes, and shipments proceed every day without any bank involvement. Answer 3 is wrong because documentation is just as essential to domestic movement: the delivery note, the transport document and the proof of delivery in this lesson served a purely domestic journey from Tema to Accra. Answer 4 confuses documents with inspection: a document describes the goods, but the carrier does not open a sealed container to read what is inside.

The takeaway: goods never travel alone. Every participant in a logistics operation acts on documents, so the documents must exist, must travel with or ahead of the goods, and must be accurate.

**Practice Question 2**

During a training session, a new employee says that an air waybill works exactly like an ocean bill of lading, because whoever holds the original document can claim the cargo at the destination airport.

Which statement best corrects this?

1. An air waybill is a negotiable document of title, but airlines keep the original instead of issuing it to the shipper.

2. An air waybill is only a quotation for the freight charges, and the real transport document is issued when the cargo arrives.

3. An air waybill acknowledges receipt and evidences the contract of carriage, but it is always non-negotiable and is not a document of title, so the cargo is delivered to the consignee named on it. ✓

4. An air waybill transfers ownership of the goods to the airline until the freight charges have been paid in full.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Both documents share their first two functions: each acknowledges receipt of the cargo and provides evidence of the contract of carriage. The difference lies in the third function. Certain forms of the ocean bill of lading function as documents of title, so possession of the properly issued original controls who may claim the goods at destination. The air waybill is always non-negotiable: it is never a document of title, and the cargo is delivered to the consignee named on it without any original being presented. That is exactly the point the new employee has misunderstood.

Answer 1 inverts the truth: negotiability is the bill of lading's feature, not the air waybill's. Answer 2 mistakes the air waybill for a quotation; the quotation-like member of the document families is the pro forma invoice, while the air waybill is a full transport document issued on receipt of the cargo. Answer 4 invents a transfer of ownership to the airline, which no transport document performs.

The takeaway: receipt and evidence of contract belong to every transport document, but the document-of-title function belongs to the ocean bill of lading alone.

**Practice Question 3**

A supermarket in Accra receives a delivery of cooking oil. The delivery note states 240 cartons, but the staff count 235 cartons before the driver leaves the premises.

Which action best describes correct professional practice?

1. Sign for the 240 cartons on the delivery note, since the document reflects what left the warehouse and the driver's paperwork must remain unchanged.

2. Record the actual count of 235 cartons and the shortage on the delivery documents before signing, so the proof of delivery shows what was really received and the discrepancy can be investigated. ✓

3. Refuse the entire delivery and send all 240 cartons back, because a shortage makes the whole consignment unacceptable.

4. Sign for 235 cartons and quietly write 235 on the driver's copy after the driver has left, so the paperwork matches without any disagreement.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

The delivery note and the proof of delivery exist to record what was actually delivered, so a shortage must be recorded on the document before signing, and the discrepancy must be reported so that both parties can investigate against evidence. Signing for 240 cartons when 235

were counted would make the receiver's own signature certify a receipt that never happened, and the parties would lose the very record the distributor needs to investigate and the supermarket may need to claim.

Answer 1 treats the document as more authoritative than the reality it is supposed to record, which defeats its purpose. Answer 3 overreacts: a shortage of five cartons, once recorded, is a claimable and investigable event, not a reason to reject the whole delivery. Answer 4 describes altering paperwork quietly after the driver has gone, which destroys the shared evidence both parties need and undermines the integrity of the record, even when the corrected figure happens to be right.

The takeaway: documents must reflect verified reality. Count, record honestly, sign, and report the discrepancy, because the proof of delivery is the evidence both sides will rely on.

### Fact-check appendix (audit notes — not for entry)

Logistics documentation, commercial invoice, packing list, bill of lading, air waybill, consignment note, delivery note, proof of delivery, shipment reference number and document discrepancy are defined in line with the lesson's key terms. — source: lesson brief 3.4, Key Terms (curriculum database).

The five purposes of logistics documentation (instructions and contracts, receipts, evidence of the right to claim goods, declarations to authorities, records supporting payment) and the principle that information moves ahead of and with the goods. — source: general business knowledge, per the lesson's drafting guidance.

The bill of lading's three classic functions: receipt for the goods, evidence of the contract of carriage, and document of title; possession of a duly endorsed bill controls the right to delivery, making it valuable for payment arrangements and for selling goods in transit. — source: research sheet trade-transport-standards.md, section 4 (ocean bill of lading).

The sea waybill is the non-negotiable alternative to the bill of lading: a receipt and evidence of contract but not a document of title. — source: research sheet trade-transport-standards.md, section 4.

The air waybill is always non-negotiable: receipt, evidence of the contract of carriage, freight bill and customs document for air cargo, but not a document of title. — source: research sheet trade-transport-standards.md, section 4.

The master air waybill is issued by the airline and the house air waybill by the freight forwarder for its part of a consolidated load. — source: research sheet trade-transport-standards.md, section 4 (master/house air waybills).

The road consignment note is non-negotiable (a receipt and evidence of the contract of carriage, not a document of title) and accompanies the goods; Europe standardises international road carriage and its consignment note under the CMR convention, Ghana is not a CMR contracting party, and West African road movements use consignment notes under national and regional practice. — source: research sheet trade-transport-standards.md, section 4 (road and rail) and DO NOT CLAIM item 4.

Certificate of origin: non-preferential certificates support ordinary (MFN) customs treatment; preferential certificates evidence qualification under a trade agreement and support claims to reduced or zero duty, for example under the ECOWAS Trade Liberalisation Scheme (ETLS) and AfCFTA certificates of origin. — source: research sheet trade-transport-standards.md, sections 5 and 12 (rules of origin).

A phytosanitary certificate covers plants and plant products and is issued by the plant-protection authority of the exporting country; a sanitary or health certificate is the counterpart for products of animal origin; quality or

inspection certificates are issued by independent inspectors attesting quality, quantity or condition at loading, with contractual (not standardised) contents. — source: research sheet trade-transport-standards.md, section 5 (trade certificates).

Customs authorities assess duties and taxes on the basis of the declared value of the goods, declared on the commercial invoice. — source: research sheet trade-transport-standards.md, section 12 (WTO valuation framework, transaction value), taught generically without rates.

In Ghana, the customs declaration for an imported consignment is lodged electronically in ICUMS (the Integrated Customs Management System), Ghana's single-window platform for trade declarations, normally by a licensed clearing agent acting for the importer, and every consignment receives a Unique Consignment Reference (UCR) for tracking. — source: research sheet ghana-trade-customs.md, sections 1 and 6 (ICUMS; clearing agents) and its SAFE PHRASINGS; no export-specific ICUMS procedure is claimed, and the worked example keeps export formalities generic.

The three-way match: the purchase order, the goods receipt/delivery record and the supplier invoice are matched before payment is released. — source: research sheet logistics-scm-frameworks.md, section 3 (procurement and the order-to-receipt flow).

A freight forwarder is an intermediary arranging international shipment for shippers, booking carriage, consolidating cargo and preparing documentation. — source: research sheet logistics-scm-frameworks.md, section 1 (freight forwarder).

Refrigerated (reefer) containers are insulated temperature-controlled equipment used for perishables, suitable for frozen shrimp. — source: research sheet trade-transport-standards.md, section 6 (containers, special types).

Receiving practice: receivers count and inspect goods against the delivery note before signing, record shortages or damage on the document at the time of receipt, and the proof of delivery (paper or electronic) evidences delivery; in modern operations electronic POD may capture signatures, photographs or system confirmations. — source: general business knowledge (operational practice), consistent with the lesson brief's definitions of delivery note and proof of delivery.

The pro forma invoice is an advance, quotation-form invoice issued before dispatch, used to confirm terms, arrange payment or apply for permits, and is not a demand for payment. — source: general business knowledge.

Shipper and consignee as the standard terms for the sender and receiver in international trade documents. — source: general business knowledge (standard trade usage).

All quantities and weights in the worked examples (500 cartons of 10 kilograms, 85 kilograms, 240 and 235 cartons) are illustrative for teaching purposes. — source: drafting instruction, WRITING_SPEC.md rule 5.1.

## Module 3 Assessment — 12 questions

*Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.*

MODULE 3 ASSESSMENT - Fundamentals of Goods Movement
Draft for entry through the instructor portal (Configure assessment step). Twelve single-choice questions, three per lesson, in lesson order. Each question carries 1 point; the platform default pass mark of 70 percent applies unless the owner changes it. All scenarios are illustrative for teaching purposes.

COVERAGE MAP
Q1 - lesson 3.1 - objective: describe the typical stages of the journey; stage ordering from collection to verification, with customs release as the gate before delivery
Q2 - lesson 3.1 - consolidation at origin and deconsolidation at destination: the cost saving, the extra handling point, and the dependence on accurate marking (objectives: explain how goods are prepared and collected; describe the role of cargo handling)
Q3 - lesson 3.1 - shipment tracking is not control: arrival opens the destination stages, and time-based charges run while the importer's side fails to act (objectives: importance of tracking and communication; factors affecting timely delivery)
Q4 - lesson 3.2 - objective: select an appropriate transportation mode for a scenario; one order split across air, rail and sea by matching each mode's speed and cost profile to the timing need
Q5 - lesson 3.2 - objective: define multimodal transport and explain how modes are combined; discriminating intermodal (one loading unit, separate contracts) from multimodal (one contract, one responsible operator)
Q6 - lesson 3.2 - objective: compare modes on accessibility; transport networks have edges, a port is never the final destination, and road performs the last mile
Q7 - lesson 3.3 - objective: distinguish general, bulk and containerised cargo; the same product changes classification with the form in which it travels
Q8 - lesson 3.3 - objective: explain the purpose of packaging; packaging must be matched to the stresses of the whole journey, not to one mode or one leg
Q9 - lesson 3.3 - objective: recognise the importance of cargo markings; identification marks tie packages to consignments and to documents at deconsolidation
Q10 - lesson 3.4 - objective: describe the basic functions of the commercial invoice and the packing list; the sale versus the boxes
Q11 - lesson 3.4 - objective: recognise the purpose of bills of lading and air waybills; the AWB-is-not-a-title-document trap in a claim-at-destination scenario
Q12 - lesson 3.4 - objectives: explain the role of proof-of-delivery records; explain why accurate and consistent documentation matters; the three-way match applied to a quantity discrepancy

**Question 1 (tests lesson 3.1)**

**1**

A consignment of machine spare parts travels from a supplier in Turkey to a buyer in Kumasi. The buyer's file lists five milestones from the journey, but the pages have fallen out of order: release of the goods by the importing country's customs; collection of the goods from the supplier; export formalities in the exporting country; verification of the goods and proof of delivery at the buyer's warehouse; the main international carriage.

Which option lists the milestones in the order the journey actually follows?

1. Collection of the goods from the supplier; the main international carriage; export formalities in the exporting country; release of the goods by the importing country's customs; verification of the goods and proof of delivery at the buyer's warehouse.

2. Collection of the goods from the supplier; export formalities in the exporting country; release of the goods by the importing country's customs; the main international carriage; verification of the goods and proof of delivery at the buyer's warehouse.

3. Collection of the goods from the supplier; export formalities in the exporting country; the main international carriage; release of the goods by the importing country's customs; verification of the goods and proof of delivery at the buyer's warehouse. ✓

4. Collection of the goods from the supplier; export formalities in the exporting country; the main international carriage; verification of the goods and proof of delivery at the buyer's warehouse; release of the goods by the importing country's customs.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

The stages of the journey run in a fixed order, and the three formal milestones hold the key positions. Goods are collected from the supplier and moved inland to the export point; the exporting country's formalities must then be completed, because the goods may not lawfully leave until the export declaration is in order. The main international carriage follows, and only on arrival do the importing country's formalities begin. Release is the gate between the arrival stages and everything that comes after: until customs grants it, containers cannot leave the terminal and consignments cannot leave a customs facility. Verification and the proof of delivery come last, closing the journey at the buyer's own premises.

Each flawed option fails on exactly one transposition. Answer 1 places export formalities after the main carriage, as if goods could sail first and be cleared to leave afterwards. Answer 2 places import release before the main carriage, although release can only follow arrival in the importing country. Answer 4 places verification before release, which would deliver goods that are still under customs control.

The takeaway: collection and clearance at origin, carriage, release at destination, then verification. Learn the gates, not just the list.

**Question 2 (tests lesson 3.1)**

**2**

A soap manufacturer in Kumasi has 9 pallets of cartoned soap for a buyer in Europe, too small to fill a container. Its freight forwarder books the pallets into a consolidated container together

with other shippers' goods. At the destination port the container is trucked to a container freight station, where the combined load is separated so that each consignee can receive its own goods.

Which option best describes what this arrangement involves?

1. The pallets travel with less handling than a full container load, so the manufacturer can use lighter packaging and fewer marks.

2. The container travels as the manufacturer's own sealed unit, because a consolidated container is reserved for one shipper until it is full.

3. Because the forwarder completes the export formalities for the combined container at origin, the goods can be delivered to the consignee abroad without import formalities or customs release.

4. The manufacturer shares the container with other shippers and pays only for the space its pallets occupy, accepting an extra handling point at each end and the need for accurate marking so that its consignment can be identified and separated at the destination. ✓

**Correct answer: Answer 4**

***Explanation***

The correct answer is Answer 4.

Consolidation combines the goods of several shippers into one container or vehicle so that each pays only for the space its goods occupy; it is the standard service for consignments too small to fill a container. The saving has a price the lesson states plainly: consolidation adds an extra handling point at each end, because the combined load must be built at origin and separated again at destination, and it depends on accurate marking, because every consignment in the combined load must be identifiable so that it can be recovered when the load is separated at the container freight station. This is the handoff idea at work: at the deconsolidation handoff, goods, marks and documents must agree, or consignments end up in the wrong pile.

Answer 1 reverses the handling truth: consolidated cargo is handled more, not less, which is why packaging and marks matter more, not less. Answer 2 misdescribes the service as the manufacturer's own sealed container, which is the full container load this small consignment is trying to avoid. Answer 3 repeats a stubborn misconception: export formalities completed at origin never replace import formalities and release in the destination country.

The takeaway: consolidation trades a freight saving for extra handling and a hard dependence on accurate marking at every handoff.

**Question 3 (tests lesson 3.1)**

**3**

An importer in Accra checks its tracking system every morning. Two weeks ago the status of its container changed to arrived at the port. Confident that the shipment is now under control, the manager waits for the goods to be delivered. No import declaration has been lodged, no collection truck has been booked, and time-based charges at the terminal have begun to accumulate.

Which statement best explains the situation?

1. The tracking system must be faulty, because a container that shows as arrived should reach the buyer's premises automatically.

2. Tracking shows the location and status of goods, but it does not lodge declarations, obtain release, or arrange collection; control comes from acting on the information, so the importer's side must now complete the formalities and book the truck. ✓

3. The shipping line is responsible for clearing and delivering imported goods, because goods in the carrier's custody remain the carrier's responsibility until they are delivered.

4. The goods must be waiting for a customs inspection that no one can influence, so the importer was right to wait.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Shipment tracking monitors the location and progress of goods; it does not itself perform any stage of the journey. The lesson's warning is exact: tracking shows location and status, but it does not clear customs, correct a wrong document, or load a truck. Control over a journey comes from what participants do with tracking information. The arrived event here is real, but arrival only opens the destination stages: the import declaration must be lodged, any duties and taxes settled, release obtained, and collection transport arranged. Because nobody acted, the goods sit after arrival, and once the allowed free period ends, storage and related charges begin to accumulate day by day.

Answer 1 blames the system, but the event is genuine; what failed is the response to it. Answer 3 confuses custody with responsibility: the carrier carries the goods, yet the import formalities belong to the importer's side, and responsibility follows the trade terms agreed in the contract rather than the physical location of the goods. Answer 4 treats delay as untouchable, but the importer's own action, a prompt declaration and a truck booked against the expected release, was exactly what the situation demanded.

The takeaway: a party who watches a tracking screen without acting on it has observation, not control.

**Question 4 (tests lesson 3.2)**

**4**

A retailer in Hamburg buys a season's stock of consumer electronics from a manufacturer in Shenzhen. The order has three parts: the launch quantity, needed at a fixed launch date; replenishment stock during the season, too time-sensitive to spend weeks at sea but not urgent enough to justify the cost of flying; and the bulk of the volume, planned months in advance and not urgent. Overland rail services connect the two regions.

Which option assigns each part of the order to the most appropriate mode?

1. Launch quantity by air, replenishment stock by rail, bulk volume by sea. ✓

2. Launch quantity by sea, replenishment stock by air, bulk volume by rail.

3. Launch quantity by rail, replenishment stock by sea, bulk volume by air.

4. All three parts by sea, because a single order should travel by a single mode at the lowest cost per unit.

**Correct answer: Answer 1**

***Explanation***

The correct answer is Answer 1.

Mode selection matches each part of the demand to the mode whose profile fits it. The launch quantity must hit a fixed date, and a missed launch forfeits sales, so it justifies air freight, the fastest and most expensive mode, used for the smallest part of the volume. The season's replenishment is the middle case the lesson describes precisely: too time-sensitive to spend weeks at sea, not urgent enough to justify flying, which is exactly where rail sits, faster than sea and cheaper than air, where the network connects the regions. The bulk of the volume is planned months in advance and not urgent, so it takes sea freight, the slowest mode but the lowest cost per unit over long distance, and that cheapness matters most at the largest quantity.

Answer 2 puts the bulk on the most expensive mode and the launch on the slowest, inverting every profile. Answer 3 is the same inversion in the other direction. Answer 4 fails because one order may sensibly travel by several modes; the lesson's comparison is a ladder on which each part of the demand finds its rung, and cost per unit is only one factor among urgency, quantity, and reliability.

The takeaway: mode selection is a portfolio decision; place each part of the order on the speed-and-cost ladder.

**Question 5 (tests lesson 3.2)**

**5**

A container is packed with goods at a factory, sealed, and carried by truck to a port. It is lifted onto a vessel, sailed to another continent, lifted onto a train at the destination port, and hauled to an inland terminal, where a truck delivers it to the warehouse. The goods are never taken out of the container. The journey was booked as separate contracts, one for each leg, with no single operator responsible for the whole movement.

Which statement best describes this arrangement?

1. It is multimodal transport, because more than one mode of transport is used.

2. It is sea freight with incidental road and rail legs, because the vessel performs the longest part of the journey.

3. It is intermodal transport, because the goods travel in the same loading unit across the changes of mode without the goods themselves being handled; the separate contracts mean it is not multimodal, which would require a single contract and one responsible operator. ✓

4. It is multimodal and intermodal at the same time, because packing the goods in one container automatically creates a single contract with one responsible operator.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

The two terms answer different questions. Intermodal transport describes how the goods physically travel: in one and the same loading unit, typically a container, across two or more modes without the goods themselves being handled at the changes of mode. Multimodal transport describes how the journey is contracted: carriage by at least two modes under a single multimodal transport contract, with one operator responsible for the goods from origin to destination. The movement described keeps the goods in one sealed container from factory to warehouse, which is intermodal in physical form, but the contracts were arranged separately for each leg, so no single operator answers for the whole journey and the multimodal definition is not met.

Answer 1 commits the most common confusion in this topic: the number of modes does not make a journey multimodal; the single contract and the single responsible operator do. Answer

2 dismisses the arrangement as one mode with extras, ignoring that a deliberate combination of modes is exactly what both terms describe. Answer 4 asserts the very equivalence the lesson denies: packing goods in a container says nothing about contracts, and the two features are independent even though both can appear in one shipment.

The takeaway: ask who is responsible and under how many contracts, and whether the goods are handled at the changes of mode; the answers, not the labels, name the arrangement.

**Question 6 (tests lesson 3.2)**

**6**

A shipping line has carried an importer's container across the ocean and discharged it at a seaport. The goods must now reach the buyer's warehouse at a specific street address in an inland city. A colleague argues that the sea freight should simply continue to the warehouse door, and that no other mode of transport is needed to complete the journey.

Which option best identifies the flaw in this reasoning?

1. There is no flaw: a booked sea freight service runs from the supplier's door to the buyer's warehouse, and the port call is only a technical pause.

2. Every mode's network has edges: the sea network runs between ports, and a port is never the final destination, so the inland leg lies outside sea freight's reach; delivery to a specific address is road freight's work, which is why road performs the last mile of almost every journey. ✓

3. The container must be unpacked at the port and the goods re-packed into railway wagons, because only rail freight is allowed to leave a port area.

4. Air freight should take over the inland leg, because it is the fastest mode and therefore the appropriate mode for every part of a journey.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Every mode runs on its own transport network, its system of routes, facilities and connections, and every network has edges. The sea network runs between ports, and the lesson states plainly that a port is never the final destination. A warehouse at a specific street address in an inland city lies outside the sea network, however good the ocean service is, so the journey cannot end with the vessel. Road freight's network is the only one that reaches ordinary addresses, which is why road carries the first and last legs of nearly every international journey and performs the last-mile delivery almost always.

Answer 1 describes a door-to-door sea service that does not exist; the quality of the booking does not extend a network. Answer 3 invents a rule about port areas: containers are routinely trucked or railed inland as whole sealed units, and no mode is confined to the port by permission. Answer 4 assigns the inland leg to air freight, whose network is the most limited of all, joining cargo airports only, and whose cost profile fits no ordinary inland delivery.

The takeaway: map the networks before costing the options; the modes that do not physically connect origin to destination are not cheaper alternatives, they are not alternatives at all.

**Question 7 (tests lesson 3.3)**

**7**

A company exports shea butter to Europe. This season it ships the butter in sealed steel drums, each drum handled, marked and counted as a unit. Next season a larger contract moves the same butter pumped directly into the heated tank of a tanker vessel, with no packaging at all.

Which statement best describes how the cargo classification changes between the two shipments?

1. The drummed butter is general cargo and the tankered butter is liquid bulk, because classification follows the form in which the goods travel, not the identity of the product. ✓

2. The classification does not change, because cargo type is a permanent property of the product.

3. Both shipments are containerised cargo, because drums and tanks are both types of containers.

4. The drummed butter is liquid bulk and the tankered butter is general cargo, because the size of the contract decides the classification.

**Correct answer: Answer 1**

***Explanation***

The correct answer is Answer 1.

Classification follows the form in which the goods travel, not the identity or the value of the product. In sealed steel drums, the butter moves as countable, packaged units, each drum handled, marked and checked individually, which is general cargo. Pumped loose into the heated tank of a tanker, the butter passes directly into the carrying compartment with no packaging at all, and nothing separates the goods from the tank; that is the definition of liquid bulk. The product has not changed, only the form in which it travels, and the classification changes with it, taking its packaging, handling equipment, storage arrangements and documents along.

Answer 2 is the beginner error the lesson's important note dismantles: cargo type is not a permanent property stamped on the product. Answer 3 misreads both forms: drums are transport packaging, not standardised freight containers, and cargo pumped into a vessel's tank is the textbook case of bulk rather than containerised cargo. Answer 4 swaps the two classes and then justifies the swap by contract size, which plays no part in classification; a larger contract changes the quantity, not the form of travel.

The takeaway: ask how the goods will travel, in packages or loose, in a hold or in a tank, and classify from that answer.

**Question 8 (tests lesson 3.3)**

**8**

A carton design has served a company well on short domestic deliveries by air within one region: dry handling, few lifts, and little stacking. The company now exports the same product in the same cartons on an ocean voyage of several weeks.

Which option best explains why the packaging may need to change?

1. Packaging is standard across all journeys, so the same cartons will perform identically at sea; only the transport documents change.

2. Only dangerous goods require packaging chosen for the journey; ordinary products may travel in any sound carton.

3. Packaging must match the whole journey: an ocean voyage brings weeks of humidity and salt-laden air, stacking beneath other cargo, and repeated lifting between truck, warehouse and ship, so the export cartons need sturdier construction and protection against moisture. ✓

4. The change is needed only because sea freight is slower than air freight, so the cartons simply need to be made heavier to survive the longer time.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Packaging serves protection and handling across the entire journey, so it must be matched to every stress the chain will apply, not to the name of the mode alone. The lesson's own contrast is exactly this one: a carton adequate for a short, dry air trip may fail on an ocean voyage of weeks through humid, salt-laden air, stacked beneath other cargo and lifted repeatedly between truck, warehouse and ship. Export packaging is therefore normally sturdier than

packaging for a short local delivery, and protection against moisture becomes part of the design rather than an optional extra.

Answer 1 is the assumption the lesson corrects: packaging is not standard across journeys, because journeys differ in what they do to the box. Answer 2 narrows journey-matched packaging to dangerous goods; regulation does impose certified packaging on dangerous goods, but every consignment's packaging must still suit its own chain, as perishables, fragile goods and ordinary exports all show. Answer 4 reduces the problem to time, yet weeks in a dry, unstacked store would harm nothing; the enemies are moisture, compression and repeated handling, and the design must answer those specific stresses.

The takeaway: matching packaging to the mode and to every handling step along the chain is a core logistics decision, not an afterthought.

**Question 9 (tests lesson 3.3)**

**9**

A consolidated container carrying several exporters' consignments arrives at a container freight station abroad, where staff unpack and separate the contents. One exporter's cartons carry no identification marks at all: no consignee code, no destination, no package numbers, only plain brown cartons. The other consignments are separated within hours; the unmarked cartons cannot be attributed to any consignment.

Which statement best identifies what went wrong and why marking matters?

1. The freight station staff were at fault, because deconsolidation should proceed without checking packages against documents.

2. The exporter's cartons lacked the identification marks that tie each package to its consignment and to the shipping documents, so at the separation point nobody could read which consignment the cartons belonged to; marks are the package's side of the conversation with the paperwork. ✓

3. The cartons should have carried dangerous goods labels instead, because only hazard labels identify whose goods a package contains.

4. Nothing was wrong with the cartons, because the container number and seal number on the outside of the container identify every package inside it.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

Cargo marking turns an anonymous box into an addressed, instructable unit. Identification marks, the consignee's code, the destination, the package number within the consignment, tie each package to its consignment and to the shipping documents, which record the same marks, counts and weights. At deconsolidation the receiver separates the combined load precisely by reading the marks against the documents: carton 137 of 1,000 is expected, counted and signed for because the documents say it exists. Where the marks are absent, the check has nothing to read, the consignment cannot be attributed, and the result is delay, misdelivery or a disputed claim. This is also why the journey lesson calls consolidation a service that depends on accurate marking.

Answer 1 blames the station's staff, but the failure was created at packing, and the staff did exactly what the process requires by checking the cargo against the documents. Answer 3 confuses the layers of marking: hazard labels carry safety information for handlers and emergency services, not consignment identity. Answer 4 fails because the container and seal numbers identify the box, not the several shippers' consignments mixed inside a consolidated load.

The takeaway: mark the packages to match the documents; a package that cannot be identified from its marks will sooner or later be handled wrongly.

**Question 10 (tests lesson 3.4)**

**10**

A receiving warehouse in Kumasi is expecting an imported consignment of 480 cartons of bottled fruit juice. Before the goods arrive, the team needs to know how the consignment is packed: how many cartons there are, how many bottles each carton contains, the gross and net weights, and how the cartons are arranged on pallets. The commercial invoice is already in the file.

Which statement best describes what the team needs?

1. Nothing further: the commercial invoice already provides all of this, because it describes the goods in detail and states the total value of the consignment.

2. The packing list, which describes the physical consignment, the packages, the quantities per package, the weights and the dimensions, and which carries no prices, because its purpose is physical rather than financial. ✓

3. The pro forma invoice, which is issued after delivery and lists the physical contents of each carton in detail.

4. The bill of lading, which itemises the contents of every carton the carrier received.

**Correct answer: Answer 2**

***Explanation***

The correct answer is Answer 2.

The two commercial-family documents describe the same consignment from two angles, and the lesson's division is sharp: the commercial invoice describes the sale, while the packing list describes the boxes. The invoice records the parties, the description, the quantities, the unit prices and the total value, and the currency; it leads payment and supports customs valuation. The packing list is deliberately physical: how many cartons, how many units each carton contains, gross and net weights, dimensions and the arrangement on pallets, and it carries no prices, because its purpose is not financial. Warehouse staff pick, count and receive against it, carriers plan loading and stowage from it, and customs compare what is physically present against it.

Answer 1 assumes the invoice covers the physical detail, but its figures serve the transaction, and nothing on it tells the team how the cartons are packed. Answer 3 misstates the pro forma invoice twice: it is issued before dispatch, as a quotation in invoice form, to confirm terms or arrange payment, not after delivery. Answer 4 gives the bill of lading a function it does not have; it acknowledges receipt of the goods and evidences the contract of carriage, and it does not itemise the contents of each carton.

The takeaway: the invoice is for the sale, the packing list is for the boxes, and the two must agree completely.

**Question 11 (tests lesson 3.4)**

**11**

A Ghanaian importer has two consignments arriving this week from the same European seller: a container of ceramic tiles by sea, travelling under a bill of lading, and a pallet of urgent machine parts by air, travelling under an air waybill. The importer's agent collects both consignments at the destination.

Which statement correctly describes how the two consignments can be claimed?

1. Both consignments are claimed by presenting the original transport document, because every transport document is a document of title.

2. Neither consignment requires any document, because goods are delivered to whoever comes to collect them.

3. The air consignment is claimed by presenting the original air waybill, which transfers title to the goods, while the sea container is delivered to the named consignee without any document.

4. The sea container is claimed with the properly issued original bill of lading, because certain forms of that document control who may claim delivery of the goods, while the air cargo is delivered to the consignee named on the air waybill without an original being presented, because the air waybill is non-negotiable and not a document of title. ✓

**Correct answer: Answer 4**

***Explanation***

The correct answer is Answer 4.

The bill of lading is the classic document serving three functions: receipt for the goods, evidence of the contract of carriage, and, for certain forms, document of title. The title function is what makes the original valuable: possession of the properly issued original controls who may claim delivery at destination, which is why goods travelling under it can even be sold while still at sea. The air waybill shares the first two functions only. It is always non-negotiable and never a document of title, so there is no original to present at the destination airport: the cargo is delivered to the consignee named on the document, and the agent's work is to complete the import formalities, not to surrender a title paper.

Answer 1 generalises the title function to every transport document, which is precisely the trap; only certain ocean bills carry it, and neither the air waybill nor the road consignment note does. Answer 2 imagines delivery without records, but every handover is made against documents. Answer 3 swaps the characters of the two documents entirely.

The takeaway: receipt and evidence of contract belong to every transport document; the document-of-title function belongs to the ocean bill of lading alone.

**Question 12 (tests lesson 3.4)**

**12**

A supermarket in Tema orders 300 cartons of rice from a distributor. Its receiving staff count 296 cartons, record the shortage on the delivery documents before signing, and the driver countersigns. A week later the distributor's commercial invoice arrives, claiming payment for 300 cartons.

Which statement best describes correct practice for the supermarket?

1. Pay the invoice in full, because the commercial invoice is the seller's formal statement of the sale and its figures govern the payment.

2. Adjust the delivery record to show 300 cartons, so that the records agree with the invoice and payment can proceed smoothly.

3. Before paying, match the order, the delivery record and the invoice; the three-way match fails for the 4 missing cartons, so payment for 300 should be held and the shortage resolved against the delivery record that both parties signed. ✓

4. Pay for 296 cartons and consider the matter closed without informing the distributor, because a payment below the invoice amount requires no explanation.

**Correct answer: Answer 3**

***Explanation***

The correct answer is Answer 3.

Before paying, a buyer matches three records: the order, the delivery record and the invoice. This three-way match exists precisely for the situation described. The order says 300, the invoice says 300, but the delivery record, honestly made at the handover and countersigned by the driver, says 296 received and 4 short. The match therefore fails for the missing cartons, and payment for 300 should be held until the shortage is resolved under the agreed terms, against the evidence both parties signed. The proof of delivery can do this job only because the discrepancy was recorded at the moment of delivery, while the driver was still present.

Answer 1 treats the invoice as governing payment, but the invoice records the sale, not the receipt; payment follows evidence of what was actually delivered. Answer 2 proposes the one thing the lesson forbids: altering the record so that the documents agree. A record changed to match the invoice is not a record but a guess with a signature on it, and the buyer loses the evidence needed to resolve the claim. Answer 4 quietly underpays and closes the matter, yet discrepancies must be reported and resolved openly, because an unexplained difference leaves both parties' records wrong.

The takeaway: sign for what you counted, match documents against each other, and let payment follow the verified record.